SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

CHAPTER: 15 **HOMEOSTASIS**

ENCIRCLE THE CORRECT CHOICE:

i) A self-regulating process by which biological systems maintain stability while adjusting to the changing conditions called
(a) Homeostasis (b) Osmoregulation (c) Excretion (d) Biological rhythms

ii) Which does not maintain a stable, homeostatic condition rather it intensifies the change that is happening to the body.
(a) Negative feedback (b) Positive feedback (c) Feedback system (d) Excretion

iii) Fresh water organisms also have specialized cells located in their gills and in skin which actively extract Na^+, Cl^- and Ca^+ from external medium and excrete (H+) or basic (HCO3-) for acid base balance in the body fluids are called
(a) Granulocyte (b) Lymphocyte (c) Ionocytes (d) podocyte

iv) Renal cortex produces an important hormone necessary for the Synthesis of RBC's:
(a) Erythropoietin (b) Leukopoietin (c) Thrombopoietin (d) Renin

v) Glomerulus have specialized cells that are wrapped around blood capillaries that play an active role in preventing plasma proteins from entering the urinary ultrafiltrate called
(a) Epithelial cells (b) Podocyte cells (c) Endothelial cells (d) None of them

vi) Tubules secretes ions such as hydrogen, potassium, and NH3 into the filtrate while reabsorbing the HCO3– from the filtrate are called
(a) Distal convoluted tubule (b) Proximal convoluted tubule (c) Collecting duct (d) Loop of Henle

vii) Glomerular filtration rate (GFR) is in which proportional to the hydrostatic pressure exerted in glomerulus wall
(a) Indirectly proportional (b) Directly proportional (c) Same proportional (d) High proportional

viii) Another compound that increases the osmotic gradient in inner medulla is the
(a) Urea (b) Water (c) Sulphates (d) Phosphates

ix) The inflammation in kidneys due to irritation of kidney stones is called
(a) Lithonephritis (b) Edema (c) Sarcoma (d) Encephalitis

x) If the surrounding environment becomes hot or a person do strenuous physical activity the heat produced inside the body and raises body temperature above suitable limit this condition is called
(a) Exothermia (b) Hyperthermia (c) Hypothermia (d) Endothermia

Q. no:	Answer	Q. no:	Answer
i)	(a) Homeostasis	vi)	(a) Distal convoluted tubule
ii)	(b) Positive feedback	vii)	(b) Directly proportional
iii)	(c) Ionocytes	viii)	(a) Urea
iv)	(a) Erythropoietin	ix)	(a) Lithonephritis
v)	(b) Podocyte cells	x)	(b) Hyperthermia

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

WRITE SHORT ANSWERS OF THE FOLLOWING QUESTIONS:

i) Why is the physiological integration of internal body environment important for living organisms?

ANS: The physiological integration of the internal body environment is of paramount importance for living organisms. It serves as the foundation for a multitude of vital functions, primarily centered around the concept of homeostasis. Homeostasis ensures that the internal environment remains stable, with factors like temperature, pH, and ion concentrations carefully regulated within narrow limits. This stability is crucial because it optimizes enzymatic reactions and metabolic processes, enabling cells, tissues, and organs to function effectively. Cells, the fundamental units of life, depend on this constancy to receive essential nutrients and oxygen, allowing them to perform their specialized functions. Enzyme activity, vital for biochemical reactions, is also contingent on maintaining the internal environment within the right parameters. Furthermore, physiological integration plays a pivotal role in regulating gas exchange, ionic balance, immune function, waste removal, and temperature regulation. Without this intricate balance, living organisms would struggle to maintain health and survival in an ever-changing external environment.

ii) Why is it important that positive and negative feedback mechanisms counter act upon each other?

ANS: It is crucial that positive and negative feedback mechanisms counteract each other because they help maintain stability and equilibrium in biological systems. Positive feedback amplifies a process, leading to change and potential instability, while negative feedback opposes change and brings the system back to its desired state. By working in opposition, these mechanisms ensure that essential physiological parameters, such as body temperature or blood glucose levels, remain within a narrow range necessary for proper functioning. For instance, in temperature regulation, positive feedback might amplify a rise in body temperature due to exercise, but negative feedback mechanisms kick in to bring it back to the optimal level. This balance is fundamental to the homeostasis of living organisms, allowing them to adapt to internal and external changes while preserving essential functions and overall health.

iii) How the aquatic Osmoregulators overcome the osmoregulatory problems?

ANS: Aquatic osmoregulators are organisms that live in water and need to maintain a stable internal osmotic environment despite the varying salt concentrations in their surroundings. They overcome osmoregulatory problems through various physiological adaptations. One key strategy is to regulate the uptake and excretion of ions and water. They may have specialized ion-transporting cells, such as chloride cells in fish gills, which actively transport ions to maintain osmotic balance. Additionally, they can adjust their urine production to regulate water and ion loss. Some aquatic osmoregulators also possess special salt-secreting glands to excrete excess salts. These mechanisms allow them to adapt to different salinity levels in their habitat, ensuring their internal balance remains stable despite changing external conditions. In essence, aquatic osmoregulators have evolved a suite of adaptive strategies to effectively manage osmoregulatory challenges in their watery environments.

iv) Why do animals excrete different types of excretory waste?

ANS: Animals excrete different types of excretory waste primarily due to variations in their physiology, diet, and habitat. The type of waste produced is closely linked to an animal's metabolic processes and the need to maintain internal homeostasis. For instance, mammals, including humans, excrete nitrogenous waste in the form of urea, which is relatively less toxic and requires less water for elimination. Birds, on the other hand, excrete uric acid, a more water-conserving method ideal for their avian lifestyle. Fish, particularly those in aquatic environments, primarily excrete ammonia, which easily diffuses into water but can be toxic in terrestrial settings. These differences in excretory waste help animals adapt to their specific ecological niches and conserve vital resources, such as water and energy, while ensuring the elimination of harmful metabolic byproducts.

v) What is the use of counter current mechanism in kidney?

ANS: The counter-current mechanism in the kidney plays a crucial role in the process of urine formation and maintaining the body's water and salt balance. This mechanism primarily occurs in the nephrons, the functional units of the kidney. It involves the flow of filtrate and blood in opposite directions within the nephron's tubules, specifically the loop of Henle. This counter-current exchange allows the kidney to concentrate urine and conserve water effectively. As filtrate moves down the descending limb of the loop of Henle, it encounters an increasing concentration of salt in the surrounding tissue, which promotes the reabsorption of water by osmosis. In contrast, as the filtrate ascends the ascending limb, it passes through an area where salt is actively pumped out, maintaining a high salt concentration in the tissue. This gradient allows for further reabsorption of salt and fine-tuning of the urine's composition. In summary, the counter-current mechanism in the kidney enables the efficient reabsorption of water and the concentration of urine, which are vital processes for maintaining proper fluid balance and regulating blood pressure in the body.

vi) What is the role of the kidney as an endocrine organ?

ANS: The kidney serves as an endocrine organ by producing and releasing hormones that play essential roles in various physiological processes. One of the key hormones produced by the kidneys is erythropoietin, which regulates the production of red blood cells in response to low oxygen levels in the blood. Additionally, the kidneys produce renin, which helps regulate blood pressure by controlling the volume of blood and the constriction of blood vessels. The conversion of vitamin D into its active form, calcitriol, also occurs in the kidneys, facilitating calcium absorption in the intestines and maintaining bone health. In summary, the kidney's endocrine function involves the secretion of hormones that contribute to blood pressure regulation, red blood cell production, and calcium homeostasis.

vii) How are kidney stones formed?

ANS: Kidney stones are formed when certain substances in the urine, such as calcium, oxalate, and uric acid, become concentrated and crystallize within the kidneys. These crystals can then bind together and gradually grow into solid masses known as kidney stones. The exact cause of kidney stone formation can vary from person to person, but several factors can contribute to their development, including genetics, diet, dehydration, and underlying medical conditions. When these stones become too large or obstruct the urinary tract, they can cause severe pain and other complications. Treatment options include medication, lifestyle changes, and, in some cases, surgical procedures to remove or break up the stones. Preventive measures like staying well-hydrated and following a balanced diet can help reduce the risk of kidney stone formation.

viii) What is the impact on the human body if kidneys suddenly or gradually lose their filtering abilities?

ANS: The loss of kidney filtering abilities, whether sudden or gradual, has a profound and detrimental impact on the body. Kidneys play a crucial role in filtering waste products and excess fluids to maintain a balanced internal environment. When this function is compromised, waste and toxins accumulate, resulting in kidney failure. Acute kidney failure can bring sudden symptoms like swelling, fatigue, and confusion, potentially becoming life-threatening if untreated. Chronic kidney disease develops slowly and may lead to high blood pressure, anemia, bone issues, and heart problems. In severe cases, when kidneys fail entirely, the only options are dialysis or kidney transplantation to replace their vital filtration function.

ix) What are the problems associated with kidney transplant?

ANS: Kidney transplant, while a life-saving procedure, is not without its challenges and potential problems. One significant issue is the risk of organ rejection, where the recipient's immune system identifies the transplanted kidney as a foreign invader and tries to destroy it. To prevent this, recipients must take immunosuppressive medications, which can have side effects and may increase the risk of infections. Additionally, finding a suitable donor can be challenging, leading to waiting lists and potential delays in receiving a transplant. Post-transplant complications can include infections, blood clots, and surgical complications. Furthermore, the long-term use of immunosuppressants can have adverse effects on other organs, like the heart and bones, and may increase the risk of certain cancers. Regular monitoring and lifelong medical management are essential for kidney transplant recipients to address these issues and ensure the success of the transplant.

x) What is hypothermia? How body overcome this condition?

ANS: Hypothermia is a medical condition characterized by a dangerously low body temperature, typically below 95 degrees Fahrenheit (35 degrees Celsius). When the body loses heat faster than it can produce it, it can lead to hypothermia. This can occur due to exposure to cold weather, cold water, or even prolonged exposure to air conditioning. To overcome hypothermia, the body employs several mechanisms. Initially, it tries to generate heat by shivering, which is a rapid muscle contraction and relaxation to produce warmth. Additionally, the body reduces blood flow to extremities to conserve heat for vital organs. It also relies on insulation through clothing or shelter to minimize heat loss. In severe cases, when these mechanisms are insufficient, medical intervention is necessary, such as rewarming the body with warm blankets or heated intravenous fluids. Hypothermia can be life-threatening, so it is essential to recognize its symptoms and take prompt action to raise the body's temperature to a safe level.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

CHAPTER: 16 **SUPPORT AND MOVEMENT**

ENCIRCLE THE CORRECT CHOICE.

i) The type of bone cell that tears down bone during the building and remodeling process are
(a) Osteocytes (b) Osteoblasts (c) Osteoclasts (d) Bone lining cells

ii) During bone formation, as the periosteum calcifies, it gives rise to a thin plate of compact bone called
(a) Periosteal bud (b) Periosteal bone collar (c) Primary ossification center (d) Epiphyseal plate

iii) The ring like shape of the girdle is due to the joining of coxal bone with the sacrum of vertebral column at anterior side and below by a joint in between pubic part called
(a) Pubic symphysis (b) Iliac junction (c) Pelvic symphysis (d) Syndesmosis

iv) Cartilage is a soft flexible form of connective tissue surrounded by a layer called
(a) Myocardium (b) periosteum (c) Osteoderm (d) Perichondrium.

v) An antagonist muscle is relaxed while the other side opposite to antagonist muscle
(a) Contracted (b) Relaxed (c) No change (d) Extremely relaxed

vi) The stage when old bone tissues are removed and gets replaced with new ones that joins both fractured sides called
(a) Bone remodeling stage (b) Bone callus formation
(c) Cartilaginous callus formation (d) Hematoma formation

vii) Which is related to the mild to severe grade of stretching or tearing of a ligament that holds the bones of a joint together?
(a) Spasm (b) Tetany (c) Sprains (d) Cramps

viii) Muscle fiber are gathered in many groups each group of fibers is called a fascicle which is surrounded by a membrane called
(a) Sarcolemma (b) Perimysium (c) Endomysium (d) Epimysium

xi) The heart muscles arranged in a branching network where cells are joined together and making a
(a) Fibrous joints (b)Sarcoplasmic reticulum (c) Gomphosis (d) T-tubules

x) These joints hold the bone by dense connective tissue contain collagenous fibers
(a) Fibrous joints (b) Syndesmosis (c) Gomphosis (d) Synovial joints

Q. no:	Answer	Q. no:	Answer
i)	(a) Homeostasis	vi)	(a) Distal convoluted tubule
ii)	(b) Positive feedback	vii)	(b) Directly proportional
iii)	(c) Ionocytes	viii)	(a) Urea
iv)	(a) Erythropoietin	ix)	(a) Lithonephritis
v)	(b) Podocyte cells	x)	(b) Hyperthermia

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

WRITE SHORT ANSWERS OF THE FOLLOWING QUESTIONS:

i) How is the cartilage useful for our skeletal system?

ANS: Cartilage is useful for our skeletal system because it serves as a flexible and supportive tissue that cushions and protects the ends of bones at joints. It helps reduce friction between bones during movement, absorbs shock, and provides a smooth surface for joint articulation.

ii) Give a comparison of major types of joints

ANS: Comparison of major types of joints:

- **Fibrous Joints:** These joints are held together by dense connective tissue, allowing little to no movement. Examples include sutures in the skull and syndesmosis in the tibia and fibula.
- **Cartilaginous Joints:** These joints are connected by cartilage and allow limited movement. Two subtypes are synchondroses (immovable, like epiphyseal plates) and symphyses (slightly movable, like intervertebral discs).
- **Synovial Joints:** These joints have a synovial cavity filled with synovial fluid, allowing for a wide range of movement. Examples include ball-and-socket joints (hip, shoulder), hinge joints (elbow, knee), and pivot joints (neck).

iii) How is the healing of fractured bone initiated pain developing?

ANS: The healing of a fractured bone is initiated by the development of pain due to the injury. Pain serves as a protective mechanism, causing the individual to immobilize the affected area. This reduced movement helps prevent further damage to the fractured bone and surrounding tissues. Pain also signals the body to begin the healing process by recruiting immune cells, initiating inflammation, and eventually forming a callus to stabilize and repair the fracture.

iv) How to conduct a first aid treatment for joint injuries and fractures?

ANS: First aid treatment for joint injuries and fractures involves immobilizing the injured area to prevent further damage. This can be done by splinting the joint or fractured bone. Apply ice to reduce swelling, and seek immediate medical attention for a proper assessment and treatment plan.

v) Give a comparison of types of muscles.

ANS: COMPARISON OF TYPES OF MUSCLES:

- **Skeletal Muscles:** Attached to bones, responsible for voluntary movements, striated appearance, and multinucleated fibers.
- **Smooth Muscles:** Found in the walls of internal organs, involuntary control, non-striated, and spindle-shaped cells.
- **Cardiac Muscles:** Compose the heart, involuntary, striated, and interconnected branched cells with intercalated discs.

vi) Why is calcium essential for cross bridge formation?

ANS: Calcium is essential for cross-bridge formation in muscle contraction because it binds to troponin, a regulatory protein on actin filaments. This binding causes a conformational change in troponin, allowing myosin (thick filament) to bind to actin (thin filament) and initiate muscle contraction.

vii) How the nervous coordination regulates muscle contraction mechanism?

ANS: Nervous coordination regulates muscle contraction by sending electrical signals (action potentials) from motor neurons to muscle fibers. These signals trigger the release of neurotransmitters (e.g., acetylcholine), which stimulate muscle fibers to contract. The nervous system controls the timing and intensity of muscle contractions.

viii) How is tetany different from tetanus?

ANS: Tetany is a condition characterized by sustained, involuntary muscle contractions due to abnormal calcium levels in the blood. Tetanus, on the other hand, is a bacterial infection caused by Clostridium tetani that leads to muscle stiffness and spasms, often resulting from a wound infection.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

ix) Explain the different types of cartilage.

ANS: Different types of cartilage include:

- **Hyaline Cartilage:** Found in joints, ribs, and respiratory passages, providing support and flexibility.
- **Elastic Cartilage:** Contains elastic fibers and is found in the ear and epiglottis, providing elasticity.
- **Fibrocartilage:** Strong and found in intervertebral discs and knee menisci, providing both support and shock absorption.

CHAPTER: 17 **NERVOUS COORDINATION**

ENCIRCLE THE CORRECT CHOICE:

i) Most of the neurons in the human brain are
(a) Sensory neurons (b) Motor neurons (c) Interneurons (d) Auditory neurons

ii) A nervous system can alter activities in its target cells in muscles and gland because:
(a) They are electrically coupled by gap junctions.
(b) The target cells have receptor proteins for the signals released by the nervous system.
(c) The nervous system releases signals into the blood to control the target cells.
(d) The target cells that become disconnected from the nervous system rapidly die.

iii) For a neuron with an initial membrane potential at -70my, an increase in the movement of potassium ions out of that neuron's cytoplasm would result in:
(a) Depolarization of the neuron (b) Hyperpolarization of neuron
(c) The replacement of potassium ions with sodium ions (d) The replacement of potassium ions with calcium ions.

iv) Action potentials move along axons
(a) More slowly in axons of large than in small diameter.
(b) The direct action of acetylcholine on the axon membrane.
(c) Activating the sodium-potassium pump at each point along the axon membrane
(d) More rapidly in myelinated than in non-myelinated axons.

v) The surface on a neuron that discharges vesicles is the:
(a) Dendrite (b) Axon hillock (c) Presynaptic membrane (d) Postsynaptic membrane.

vi) An inhibitory postsynaptic potential occurs in a membrane made more permeable to:
(a) Potassium ions (b) Sodium ions (c) Calcium ions (d) ATP

vii) The major inhibitory neurotransmitter of the brain is:
(a) Acetylcholine (b) Epinephrine (c) GABA (d) Endorphin

viii) Where are neurotransmitter receptors:
(a) On the nuclear membrane located (b) In the myelin sheet
(c) At nodes of Ranvier (d) On the postsynaptic membrane

ix) Which selection is incorrectly paired
(a) Forebrain → Diencephalon (b) Fore brain → Cerebrum
(c) Midbrain → Brainstem (d) Midbrain → Cerebellum

x) Which of the following receptors is incorrectly paired with the type of energy is transduces?
(a) Mechanoreceptors → Sound (b) Pain receptors→ Electricity -
(c) Chemoreceptors → Solute concentration (d) Thermo receptors →Heat

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

Q. no:	Answer	Q. no:	Answer
i)	(c) Interneurons	vi)	(a) Potassium ions
ii)	(b) The target cells have receptor proteins for the signals released by the nervous system.	vii)	(c) GABA
iii)	(b) Hyperpolarization of neuron	viii)	(d) On the postsynaptic membrane
iv)	(d) More rapidly in myelinated than in non-myelinated axons.	ix)	(d) Midbrain → Cerebellum
v)	(c) Presynaptic membrane	x)	(b) Pain receptors → Electricity

WRITE SHORT ANSWERS OF THE FOLLOWING QUESTIONS:

i) Why receptors are called transducers?

ANS: Receptors are called transducers because they serve as specialized structures or cells in the sensory organs that convert one form of energy or signal into another. Specifically, they transform external stimuli (such as light, sound, pressure, temperature, or chemicals) into electrical signals. This process of transduction is essential for the nervous system to interpret and respond to sensory information effectively.

ii) Why neurolemma is polarized?

ANS: The neurolemma, also known as the Schwann cell membrane, is polarized because it has a distinct separation of electrical charges across its membrane. This polarization is crucial for the proper functioning of neurons and the conduction of nerve impulses. The neurolemma has a resting membrane potential, which means there is a difference in electrical charge (voltage) between the inside and outside of the cell membrane. This polarization is established and maintained by ion channels, pumps, and selective permeability to ions, particularly sodium (Na^+) and potassium (K^+). It allows the neurolemma to rapidly change its membrane potential in response to nerve signals, facilitating the propagation of action potentials along the axon.

iii) What do you mean by threshold potential?

ANS: The threshold potential is the critical level of depolarization that a neuron's membrane must reach to generate an action potential (nerve impulse). In other words, it is the minimum membrane potential required to trigger the opening of voltage-gated sodium channels and initiate an action potential. Once the threshold potential is reached (typically around -55 to -50 millivolts in neurons), it triggers a rapid and self-propagating change in membrane potential, resulting in the transmission of the nerve impulse along the neuron's axon. Below the threshold potential, no action potential is generated, as the sodium channels remain closed, and the neuron remains at its resting state.

iv) Why impulses move faster in myelinated neurons?

ANS: Impulses move faster in myelinated neurons due to a process called "saltatory conduction." In myelinated neurons, the axon is covered by segments of a fatty substance called myelin, which acts as an insulating sheath. The myelin sheath is interrupted by small gaps called "Nodes of Ranvier."

Saltatory conduction allows impulses to "jump" from one Node of Ranvier to the next, rather than propagating continuously along the entire length of the axon. This jumping or "leaping" of the action potential significantly increases the speed of nerve impulse transmission. Here's why:

- **Energy Conservation:** Saltatory conduction conserves energy because the action potential is only regenerated at the Nodes of Ranvier, where ion channels are concentrated. This reduces the energy required for maintaining the membrane potential along the entire length of the axon.
- **Increased Speed:** The myelin sheath insulates the axon, preventing the loss of ions across the membrane. As a result, the action potential can travel rapidly from one node to the next, effectively "skipping" the myelinated regions. This allows for faster conduction of the nerve impulse.
- **Efficient Transmission:** Myelination and saltatory conduction are especially advantageous for long axons because they enable the rapid transmission of impulses over long distances without significant loss of signal strength.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

v) Why interneurons are called association neuron?

ANS: Interneurons are often referred to as association neurons because they play a crucial role in connecting and integrating information between sensory neurons and motor neurons within the central nervous system (CNS). They "associate" or relay signals between different neurons, helping process and interpret sensory information, make decisions, and initiate appropriate motor responses. Interneurons are responsible for complex neural functions, including memory, learning, and decision-making, making them vital for higher-order cognitive processes.

vi) What do you mean by saltatory conduction?

ANS: Saltatory conduction is a method of nerve impulse propagation that occurs in myelinated neurons. It involves the rapid and "leaping" transmission of action potentials (nerve impulses) along the axon, specifically from one Node of Ranvier to another. The myelin sheath insulates the axon except at these nodes, where ion channels are concentrated. Action potentials are generated at the nodes and "jump" quickly to the next node, bypassing the myelinated regions. This mode of conduction significantly increases the speed and efficiency of nerve impulse transmission along the axon. Saltatory conduction is essential for the rapid communication of nerve signals over long distances in the nervous system while conserving energy.

vii) Differentiate between following:

ANS: (a) ELECTRICAL VS. CHEMICAL SYNAPSE:

ASPECT	ELECTRICAL SYNAPSE	CHEMICAL SYNAPSE
NATURE OF SIGNAL TRANSFER	Direct electrical connection	Indirect chemical transmission
SPEED OF TRANSMISSION	Extremely fast	Relatively slower
SYNAPTIC CLEFT	Absent	Present
NEUROTRANSMITTER	None	Neurotransmitters are involved
BIDIRECTIONAL TRANSMISSION	Yes	Typically unidirectional
COMMON LOCATIONS	Cardiac muscle, some areas of CNS	Most synapses in the nervous system

(b) CNS vs. PNS:

ASPECT	CENTRAL NERVOUS SYSTEM (CNS)	PERIPHERAL NERVOUS SYSTEM (PNS)
LOCATION	Brain and spinal cord	Nerves outside the brain and spinal cord
FUNCTION	Integration of information, decision-making, thought processes, and higher functions	Transmission of information between the CNS and the rest of the body, including sensory and motor functions
COMPONENTS	Brain and spinal cord	Nerves (sensory and motor neurons), ganglia, and receptors
SUBDIVISIONS	Brain (cerebrum, cerebellum, etc.) and spinal cord	Autonomic (sympathetic and parasympathetic) and somatic nervous systems
CONTROL OVER	Overall control of bodily functions	Facilitates communication with the CNS and regulates specific bodily functions
PROTECTION	Protected by the skull and vertebral column	Not as heavily protected, more exposed to external influences

(c) EXCITATORY VS. INHIBITORY NEUROTRANSMITTERS:

ASPECT	EXCITATORY NEUROTRANSMITTERS	INHIBITORY NEUROTRANSMITTERS
EFFECT ON POSTSYNAPTIC CELL	Promote depolarization	Inhibit depolarization
MEMBRANE POTENTIAL CHANGE	Increases membrane potential, making it more likely for the postsynaptic neuron to fire an action potential	Decreases membrane potential, making it less likely for the postsynaptic neuron to fire an action potential
EXAMPLES	Glutamate, serotonin, dopamine	GABA (Gamma-Aminobutyric Acid), glycine
FUNCTION	Enhance neural communication by exciting the postsynaptic cell	Regulate neural communication by inhibiting the postsynaptic cell

ROLE IN NEURAL PROCESSING	Essential for signal propagation and information transmission	Help control and fine-tune neural signaling

(d) PARASYMPATHETIC VS. SYMPATHETIC NERVOUS SYSTEM:

ASPECT	PARASYMPATHETIC NERVOUS SYSTEM	SYMPATHETIC NERVOUS SYSTEM
ACTIVATION EFFECTS	Promotes "rest and digest" response, conserves energy	Initiates "fight or flight" response, prepares the body for action
PREGANGLIONIC NEUROTRANSMITTER	Acetylcholine	Acetylcholine
POSTGANGLIONIC NEUROTRANSMITTER	Acetylcholine	Norepinephrine
LOCATION OF GANGLIA	Near or within target organs	Close to the spinal cord, farther from target organs
EFFECT ON HEART RATE	Slows down (bradycardia)	Speeds up (tachycardia)
EFFECT ON PUPILS	Constricts (miosis)	Dilates (mydriasis)
EFFECT ON DIGESTIVE SYSTEM	Stimulates digestion and peristalsis	Inhibits digestion and peristalsis
STRESS RESPONSE	Counteracts stress responses	Enhances stress responses
DOMINANT IN RESTING STATE	Yes	No (sympathetic is dominant)

viii) Define following terms: (a) Endorphins (b) Pacinian corpusices (c) Nociceptors (d) Reflex arc (e) Stroke (f) Meningitis

ANS: DEFINITIONS:

(a) ENDORPHINS: Endorphins are natural neurotransmitters produced by the body that act as natural painkillers and mood elevators. They are often released in response to stress and pain and can induce feelings of well-being and reduce pain perception.

(b) PACINIAN CORPUSCLES: Pacinian corpuscles are specialized sensory receptors found in the skin and other tissues. They are responsible for detecting mechanical pressure and vibration.

(c) NOCICEPTORS: Nociceptors are sensory receptors that respond to potentially harmful stimuli, such as tissue damage or extreme temperature, and transmit signals that are perceived as pain by the brain.

(d) REFLEX ARC: A reflex arc is a neural pathway that controls a reflex action. It typically includes a sensory receptor, an afferent (sensory) neuron, an interneuron (in some cases), an efferent (motor) neuron, and an effector organ or muscle. Reflex arcs allow for rapid, involuntary responses to specific stimuli.

(e) STROKE: A stroke is a medical condition characterized by the sudden loss of blood flow to a part of the brain, leading to brain cell damage or death. It can result in various neurological symptoms, including paralysis, speech difficulties, and cognitive impairment.

(f) MENINGITIS: Meningitis is the inflammation of the meninges, which are the protective membranes surrounding the brain and spinal cord. It can be caused by infections (viral or bacterial) and can lead to symptoms such as severe headaches, fever, and neck stiffness.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

CHAPTER: 18 CHEMICAL COORDINATION

ENCIRCLE THE CORRECT CHOICE:

i) Which of the following about hormones is incorrect statements?

(a) They are produced by endocrine glands.
(b) They are modified amino acids, peptides or steroid molecules.
(c) They are carried by the circulatory system.
(d) They are used to communicate between different organisms.

ii) Oxytocin and ADH are produced by the and stored in the

(a) Hypothalamus; Neurohypophysis (b) Adenohypophysis; Kidneys
(c) Adrenal cortex; Adrenal medulla (d) Posterior pituitary; Anterior pituitary

iii) Tropic hormones from the anterior pituitary directly affect the release of which of the following?

(a) Parathyroid hormone (b) Calcitonin (c) Epinephrine (d) Thyroxine

iv) Which of the following endocrine disorders is/are not correctly matched with the malfunctioning gland?

(a) Diabetes → Pancreas (b) Giantism→ Pituitary (c) Goiter Adrenal medulla (d) Tetany Parathyroid

v) Which hormone exerts antagonistic action to PTH?

(a) Thyroxin (b) Calcitonin (c) Growth hormone (d) Epinephrine

vi) Which of the following are synthesized from the amino acid tyrosine?

(a) Epinephrine (b) Catecholamines (c) Thyroxin (d) All of the following

vii) If the adrenal cortex were removed, which group of hormones would be most affected?

(a) Steroid (b) Peptide (c) Tropic (d) Amino acid derived

vii) Which of the following hormones is (are) secreted by the adrenal gland in response to stress and promote(s) the synthesis of glucose from non-carbohydrate substrates?

(a) Glucagon (b) Glucocorticoids (c) Epinephrine (d) Thyroxin

ix) Which of the following stimulates the contraction of uterine muscle?

(a) Oxytocin (b) Thyroxin (c) Growth hormone (d) Insulin

x) A distinctive feature of the mechanism of action of thyroid hormones and steroid hormones is that

(a) These hormones are regulated by feedback loop
(b) These hormones bind with specific receptor protein on the plasma membrane of target cells.
(c) These hormones bind to receptors inside cells
(d) Target cells react more rapidly to these hormones than to local regulators.

Q. no:	Answer	Q. no:	Answer
i)	(d) They are used to communicate between different organisms.	vi)	(d) All of the following
ii)	(a) Hypothalamus; Neurohypophysis	vii)	(a) Steroid
iii)	(d) Thyroxine	viii)	(b) Glucocorticoids
iv)	(c) Goiter Adrenal medulla	ix)	(a) Oxytocin
v)	(b) Calcitonin	x)	(c) These hormones bind to receptors inside cells

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

WRITE SHORT ANSWERS OF THE FOLLOWING QUESTIONS:

i) Why hydrophilic hormones need secondary messengers?

ANS: Hydrophilic hormones need secondary messengers because they cannot easily cross the cell membrane due to their water-soluble nature. These hormones, such as peptide hormones and some catecholamines, bind to receptors on the cell surface and trigger a cascade of intracellular events through secondary messengers like cyclic AMP (cAMP) or calcium ions. Secondary messengers relay the hormonal signal from the cell surface to the cell's interior, leading to cellular responses.

ii) Why hormones are called chemical messengers?

ANS: Hormones are called chemical messengers because they are signaling molecules produced by endocrine glands and released into the bloodstream. They travel throughout the body and interact with specific target cells or tissues to regulate various physiological processes. Hormones convey information and instructions to these target cells, coordinating activities and maintaining homeostasis within the body.

iii) Why posterior pituitary is called neurohypophysis?

ANS: The posterior pituitary is called the neurohypophysis because it is an extension of the hypothalamus and is primarily composed of nerve fibers. It stores and releases hormones, oxytocin and vasopressin (antidiuretic hormone), which are produced by the hypothalamus. These hormones are transported down the nerve fibers from the hypothalamus to the posterior pituitary, where they are released into the bloodstream.

iv) Enlist hormones of anterior pituitary?

ANS: Hormones of the anterior pituitary gland include:

i) Growth Hormone (GH)
ii) Adrenocorticotropic Hormone (ACTH)
iii) Thyroid-Stimulating Hormone (TSH)
iv) Follicle-Stimulating Hormone (FSH)
v) Luteinizing Hormone (LH)
vi) Prolactin (PRL)

v) Why pancreas is called as exocrine and endocrine gland?

ANS: The pancreas is called both an exocrine and endocrine gland because it has dual functions:

i) **Exocrine function:** The pancreas secretes digestive enzymes into the pancreatic ducts, which eventually reach the small intestine. These enzymes aid in the digestion of carbohydrates, proteins, and fats.

ii) **Endocrine function:** The pancreas also contains clusters of cells called the islets of Langerhans, which secrete hormones directly into the bloodstream. The main endocrine hormones produced by the pancreas are insulin (produced by beta cells) and glucagon (produced by alpha cells), which regulate blood glucose levels.

vi) Why secretions of adrenal cortex are called corticosteroids?

ANS: The secretions of the adrenal cortex are called corticosteroids because they are steroid hormones produced by the adrenal cortex. These hormones include glucocorticoids (such as cortisol), mineralocorticoids (such as aldosterone), and small amounts of sex hormones (such as androgens). They play essential roles in regulating metabolism, immune response, electrolyte balance, and other physiological processes.

vii) Define following terms

(a) Hormones **(b) Goiter** **(c) Thymosin**
(d) GnTH **(e) Atrial natriuretic hormone** **(f) Erythropoietin**

ANS: DEFINITIONS:

(a) Hormones: Hormones are chemical messengers produced by glands and tissues in the body. They regulate various physiological processes by binding to specific receptors on target cells and influencing cellular functions.

(b) Goiter: Goiter is an enlargement of the thyroid gland, typically caused by an iodine deficiency or thyroid dysfunction. It can lead to swelling in the neck and disrupt normal thyroid hormone production.

(c) Thymosin: Thymosin is a hormone produced by the thymus gland, and it plays a role in the development and maturation of T lymphocytes (a type of white blood cell) important for the immune system.

(d) GnRH (Gonadotropin-Releasing Hormone): GnRH is a hormone produced by the hypothalamus that regulates the release of gonadotropins (luteinizing hormone and follicle-stimulating hormone) from the anterior pituitary, which in turn control the functions of the gonads (testes in males and ovaries in females).

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(e) Atrial Natriuretic Hormone (ANH): ANH is a hormone produced by the atria of the heart in response to increased blood volume and pressure. It helps regulate blood pressure by promoting the excretion of sodium and water by the kidneys.

(f) Erythropoietin: Erythropoietin (EPO) is a hormone produced by the kidneys in response to low oxygen levels in the blood. It stimulates the production of red blood cells in the bone marrow, increasing oxygen-carrying capacity.

viii) Differentiate between following:

(a) Hypothyroidism and Hyperthyroidism **(b) Insulin and glucagon**

(c) Calcitonin and Parathyroid hormone **(d) Lipophilic and hydrophilic hormones**

ANS: DIFFERENTIATE BETWEEN

(A) HYPOTHYROIDISM AND HYPERTHYROIDISM

ASPECT	HYPOTHYROIDISM	HYPERTHYROIDISM
Thyroid Hormone Production	Insufficient production of thyroid hormones	Excessive production of thyroid hormones
Symptoms	Fatigue, weight gain, cold intolerance	Weight loss, increased heart rate, heat intolerance
Causes	Often due to autoimmune disease (Hashimoto's)	Often due to autoimmune disease (Graves' disease)

(b) INSULIN AND GLUCAGON

ASPECT	INSULIN	GLUCAGON
Function	Lowers blood glucose levels	Raises blood glucose levels
Source	Produced by pancreatic beta cells	Produced by pancreatic alpha cells
Effects on Cells	Promotes glucose uptake by cells	Stimulates glucose release from liver

(c) CALCITONIN AND PARATHYROID HORMONE

ASPECT	CALCITONIN	PARATHYROID HORMONE (PTH)
Source	Produced by thyroid gland	Produced by parathyroid glands
Function	Lowers blood calcium levels	Raises blood calcium levels
Target Tissues	Bone cells	Bone, kidneys, and intestines

(d) LIPOPHILIC AND HYDROPHILIC HORMONES

ASPECT	LIPOPHILIC HORMONES	HYDROPHILIC HORMONES
Solubility	Lipid-soluble	Water-soluble
Receptor Location	Intracellular receptors	Cell surface receptors
Example	Steroid hormones (e.g., cortisol)	Peptide hormones (e.g., insulin)

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

CHAPTER: 19 **ANIMAL BEHAVIOUR**

ENCIRCLE THE CORRECT CHOICE:

i) The decrease in response to repeated or continuous stimulation is called:
(a) Insight (b) Habituation (c) Maturation (d) Instinct

ii) The cyclic physiological patterns of activities in an organism that are in response to periodic environmental changes is called
(a) Biological Clock (b) Biological rhythms (c) Circadian rhythms (d) All of Above

iii) A directional movement of an organism toward or away from the particular stimulus is called
(a) Kinesis (b) Taxis (c) Reflex (d) Fixed Action Pattern

iv) Simplest form of learning, in which the animals learn not to respond or ignore to a repeated, irrelevant stimulus.
(a) Imprinting (b) Kinesis (c) Habituation (d) Fixed Action Pattern

v) Which one of the following is not an instinct behaviour?
(a) Migration of Salmon Fish (b) Dances of bees (c) Territorial behaviour in Gorillas (d) Nest building by birds

vi) Wolfgang Kohler in 1920s performed the experiment on the behaviour of
(a) Rats (b) Chimpanzees (c) Graylag Goose (d) Dog

vii) Workers (honey bee) are
(a) Fertile Females (b) Fertile Male (c) Sterile Female (d) Sterile Male

viii) Agonistic behaviour is the type of
(a) Innate Behaviour (b) Learning behaviour (c) Social Behaviour (d) Instinctive behaviour

ix) Migration of Salmon Fish is the example of
(a) Orientation behaviour (b) Learning behaviour (c) Social Behaviour (d) Instinctive behaviour

x) Skinner box was used for
(a) Classic Conditioning (b) Operant Learning (c) Latent Learning (d) Insight Learning

Q. no:	Answer	Q. no:	Answer
i)	(b) Habituation	vi)	(b) Chimpanzees
ii)	(b) Biological rhythms	vii)	(c) Sterile Female
iii)	(b) Taxis	viii)	(c) Social Behaviour
iv)	(c) Habituation	ix)	(d) Instinctive behaviour
v)	(c) Territorial behaviour in Gorillas	x)	(b) Operant Learning

WRITE SHORT ANSWER OF THE FOLLOWING QUESTIONS:

i) Define stimuli. How organisms respond to different stimuli?

ANS: Stimuli are external or internal changes in an organism's environment that trigger a response. Organisms respond to different stimuli through their sensory organs, which detect and transmit information about the stimuli to the organism's nervous system. The nervous system processes this information and generates a suitable response, which can be a behavioral, physiological, or even a cellular response. How organisms respond to stimuli depends on their sensory abilities, their internal state, and their genetic programming.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

ii) What is reflex? Give three examples of reflexes in vertebrates.

ANS: A reflex is an involuntary and rapid response to a specific stimulus. Three examples of reflexes in vertebrates are:

- **Knee-jerk reflex:** When a doctor taps the tendon below the kneecap with a hammer, the leg kicks forward involuntarily.
- **Blink reflex:** When a foreign object or sudden bright light approaches the eye, the eyelids close automatically to protect the eye.
- **Withdrawal reflex:** Touching a hot object causes an immediate withdrawal of the hand or limb from the source of heat to prevent injury.

iii) What is the relationship between heredity and behaviour?

ANS: The relationship between heredity and behavior lies in the influence of genetics on an organism's behavioral traits. Heredity, through genes and genetic inheritance, can predispose individuals to certain behavioral tendencies. These tendencies can include instincts, preferences, or susceptibilities to certain behaviors. However, it's important to note that behavior is also influenced by environmental factors and learning, so the interaction between genes and the environment plays a significant role in shaping behavior.

iv) Define biological rhythms. How are biological rhythms important to man?

ANS: BIOLOGICAL RHYTHMS:

Biological rhythms refer to the cyclical and predictable patterns of physiological or behavioral activities exhibited by organisms in response to periodic environmental changes. These rhythms can be driven by internal biological clocks or external cues, such as light-dark cycles or temperature changes. Biological rhythms can occur on various timescales, including daily (circadian rhythms), monthly (lunar rhythms), and yearly (annual rhythms).

Biological rhythms are important to humans for several reasons:

i. **Regulation of Sleep-Wake Cycles:** Circadian rhythms synchronize our internal body clocks with day and night, impacting sleep patterns and disruptions due to factors like shift work or jet lag.
ii. **Hormone Production:** Hormones like cortisol, melatonin, and growth hormone follow daily or seasonal rhythms, influencing overall health, metabolism, and the immune system.
iii. **Body Temperature Regulation:** Circadian rhythms affect body temperature, with the lowest at night and highest in the afternoon, influencing alertness and physical performance.
iv. **Reproductive and Fertility Cycles:** Menstrual and reproductive cycles are influenced by biological rhythms, crucial for family planning and fertility treatments.
v. **Behavioral Patterns:** Biological rhythms affect behavioral patterns, such as eating and activity schedules, even in animals aligning their daily activities with circadian rhythms.
vi. **Health and Disease:** Disruptions in biological rhythms are linked to sleep disorders, mood disorders, metabolic issues, and cardiovascular diseases, aiding research and treatment development.
vii. **Optimal Timing for Activities:** Knowledge of these rhythms helps optimize timing for activities like exercise and medication, with certain activities being more effective at specific times of the day.

v) What do you know about taxis? Give examples of positive taxis and negative taxis.

ANS: Taxis is a behavioral response in which an organism moves either toward (positive taxis) or away from (negative taxis) a stimulus. Examples of positive taxis include:

i. **Phototaxis:** Movement toward light by plants or organisms like moths.
ii. **Chemotaxis:** Bacterial movement toward a chemical gradient, such as a nutrient source.

Examples of negative taxis include:

i. **Geotaxis:** Movement away from the gravitational pull, such as plant roots growing away from gravity.
ii. **Hydrotaxis:** Movement away from excessive moisture.

vi) What is the difference between innate behaviour and learning behaviour?

ANS: DIFFERENCE BETWEEN INNATE BEHAVIOR AND LEARNING BEHAVIOR:

ASPECT	INNATE BEHAVIOR	LEARNING BEHAVIOR
DEFINITION	Genetically programmed responses	Acquired through experience or teaching
DEVELOPMENT	Present from birth or hatching	Developed over time with exposure
DEPENDENCY ON	Not dependent on prior experience	Heavily dependent on prior

EXPERIENCE		experience
FLEXIBILITY	Less flexible, stereotypical	More flexible and adaptable
EXAMPLES	Reflexes, instincts, fixed action patterns	Language, problem-solving, skills

vii) How do bees communicate about food sources?

ANS: Bees communicate about food sources primarily through a dance known as the "waggle dance." This dance conveys information about the location of a food source to other members of the hive. Here's how it works:

i. **Scout Discovery:** When a scout bee discovers a food source, it returns to the hive.
ii. **Waggle Dance:** The scout performs a waggle dance on the vertical comb within the hive. The dance consists of a figure-eight pattern, with the direction and duration of the waggle part indicating the location of the food source.
   - **Direction:** The angle of the waggle relative to the vertical axis of the comb indicates the direction of the food source relative to the position of the sun.
   - **Duration:** The length of time the waggle dance lasts conveys the distance to the food source. Longer dances indicate greater distances.
iii. **Recruitment:** Other worker bees observe the dance and interpret the information. They use this information to navigate to the food source, collecting nectar, pollen, or other resources.

This sophisticated form of communication allows bees to efficiently share information about the location of valuable food sources.

viii) How do old word monkeys defend their territory?

ANS: Old World monkeys, which include species like baboons and macaques, use various aggressive and territorial behaviors to defend their territory. These behaviors help them maintain access to critical resources and protect their group. Some of the strategies they employ include:

i. **Vocalizations:** Old World monkeys often use vocalizations, such as loud calls, roars, or alarm calls, to signal their presence and warn intruders.
ii. **Physical Displays:** Monkeys may engage in physical displays like posturing, raising their hackles, or showing their teeth to intimidate potential intruders.
iii. **Chasing and Aggression:** In some cases, Old World monkeys may engage in physical confrontations with rival monkeys. They may chase or fight with intruders to establish dominance and protect their territory.
iv. **Group Coordination:** These monkeys often live in social groups, and they may cooperate to defend their territory. Group members can collectively deter or confront intruders.
v. **Patrolling:** Some Old World monkey species engage in territorial patrolling, where they actively move through their territory, ensuring it remains free of intruders.

These territorial behaviors are essential for maintaining group cohesion, protecting resources like food and shelter, and ensuring the survival and reproductive success of the group.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

CHAPTER: 20 **REPRODUCTION**

ENCIRCLE THE CORRECT CHOICE:

i) Ovarian cycle includes changes in ovarian follicles to secondary follicle and finally to
(a) Graafian follicle (b) Primary follicle (c) Vesicular follicle (d) Both "a" and "c"

ii) In males, FSH promotes the development and maturation of?
(a) Testosterone (b) Sperm cells (c) Egg cells (d) Polar cells

iii) The menstrual cycle is primarily regulated by all of the following except:
(a) FSH (b) Estrogen (c) Androgen (d) Progesterone

iv) FSH induces the Graafian follicle to produce
(a) LH (b) Estrogen (c) Androgen (d) Progesterone

v) Key hormones in female reproductive system that regulates menstrual cycle.
I. Estrogen **II. Progesterone** **III. Testosterone**
(a) I only (b) II only (c) I and II (d) III only

vi) What is the medical term for a miscarriage?
(a) Spontaneous abortion (b) Induced abortion (c) Ectopic pregnancy (d) Both "a" and "b"

vii) Syphilis is a sexually transmitted disease caused by:
(a) Bacteria (b) Virus (c) Protozoan (d) Yeast

viii) Structure that protects the testes and keeps them at a temperature several degrees below the normal body temperature.
(a) Epididymis (b) Scrotum (c) Penis (d) All of the above

ix) In-vitro fertilization (IVF) is a method used to:
(a) Treat infertility (b) Enhance hormones (c) Prevent STDs (d) None of the above

x) Which of the following is a potential cause of female infertility?
(a) Polycystic ovary syndrome (b) Endometriosis (c) Fallopian tube blockage (d) All of these

Q. no:	Answer	Q. no:	Answer
i)	(a) Graafian follicle	vi)	(a) Spontaneous abortion
ii)	(b) Sperm cells	vii)	(a) Bacteria
iii)	(c) Androgen	viii)	(b) Scrotum
iv)	(b) Estrogen	ix)	(a) Treat infertility
v)	(c) I and II	x)	(d) All of these

WRITE SHORT ANSWER OF THE FOLLOWING QUESTIONS:

i) Why urethra in male is called urinogenital duct?
ANS: The urethra in males is called the urinogenital duct because it serves as a common passageway for both urine and genital (reproductive) fluids. It plays a dual role in the male reproductive and urinary systems.
i). Urinary Function: The urethra carries urine from the urinary bladder and expels it from the body during urination. This is the primary function of the urethra in both males and females.
ii). Genital Function: In males, the urethra also serves a reproductive function. It acts as a conduit for the transport of seminal fluids, including sperm, from the male reproductive organs (such as the epididymis and seminal vesicles) to

the external environment during ejaculation. This allows for the release of sperm and other fluids during sexual intercourse for the purpose of fertilization.

So, the term "urinogenital duct" reflects the dual role of the male urethra in carrying both urine and reproductive fluids, making it an important structure in the male reproductive system as well as the urinary system.

ii) Why hormonal system of female is better than male?

ANS: It is not accurate to say that the hormonal system of females is inherently "better" than males. Both males and females have unique hormonal systems that serve specific functions in their respective reproductive systems and overall health. However, it can be noted that the female hormonal system is more complex and undergoes more significant cyclic changes due to the menstrual cycle and pregnancy, whereas the male hormonal system tends to be more stable throughout life.

iii) Why testes descend down in human male foetus before birth?

ANS: Testes descend down in human male fetuses before birth due to a process called testicular descent or inguinal descent. This is essential for proper testicular development and function. The testes initially develop within the abdominal cavity near the kidneys. As the fetus matures, the testes gradually descend through the inguinal canal into the scrotum. This descent is necessary because the lower temperature in the scrotum is crucial for spermatogenesis (sperm production) and overall male reproductive health.

iv) What are symptoms of gonorrhea?

ANS: SYMPTOMS OF GONORRHEA:

Gonorrhea is a sexually transmitted infection (STI) caused by the bacterium Neisseria gonorrhoeae. The symptoms of gonorrhea can vary among individuals, and some people with the infection may not experience noticeable symptoms at all, especially in the early stages. When symptoms do occur, they typically manifest within a few days to a couple of weeks after exposure. Common symptoms of gonorrhea include:

i. **Painful Urination:** A burning sensation while urinating is a frequent symptom of gonorrhea.
ii. **Unusual Genital Discharge:** Both males and females may experience abnormal discharge from the genitals. In males, this discharge is often white, yellow, or greenish and may be thick or thin. In females, the discharge may be more subtle or mistaken for a mild vaginal infection.
iii. **Pelvic or Genital Pain:** Some individuals may experience discomfort or pain in the lower abdomen, pelvis, or genitals.
iv. **Rectal Symptoms:** If the infection is contracted through anal intercourse, it can lead to symptoms such as anal discharge, discomfort, or bleeding.
v. **Sore Throat:** Gonorrhea can infect the throat if contracted through oral sex, leading to a sore throat.
vi. **Swollen Testicles:** In rare cases, gonorrhea can cause inflammation and pain in the testicles.

v) Why inner uterine wall thicken during secretory phase?

ANS: The inner uterine wall, also known as the endometrium, undergoes cyclic changes during the menstrual cycle, including thickening during the secretory phase. The secretory phase is the second half of the menstrual cycle, occurring after ovulation (the release of an egg from the ovary). Here's why the inner uterine wall thickens during this phase:

i) PREPARATION FOR IMPLANTATION: The primary purpose of the thickening of the endometrium is to prepare it for potential embryo implantation. After ovulation, if fertilization of the egg occurs, the resulting embryo needs a suitable environment to implant and develop in the uterus. The thickened endometrium provides this nourishing and supportive environment.

ii) Hormonal Changes: Hormones play a crucial role in regulating the menstrual cycle and the changes in the endometrium. After ovulation, the corpus luteum (a temporary endocrine structure formed from the ovarian follicle after ovulation) produces progesterone. Progesterone stimulates the endometrium to thicken, increase its blood supply, and develop secretory glands. These changes make the endometrium more receptive to a fertilized egg.

iii) MAINTENANCE OF PREGNANCY: If fertilization occurs, the thickened endometrium continues to provide essential nutrients and support to the developing embryo during its early stages. This is critical for the establishment and maintenance of a pregnancy.

iv) CYCLIC SHEDDING: If fertilization does not occur, the thickened endometrial lining will eventually shed during menstruation. This shedding is a natural part of the menstrual cycle and marks the beginning of a new cycle.

vi) Enlist causes of male and female infertility.

ANS: CAUSES OF MALE INFERTILITY:

i. **Abnormal Sperm Production or Function:** This includes conditions like oligospermia (low sperm count) and azoospermia (absence of sperm in the ejaculate).
ii. **Ejaculation Disorders:** Problems with ejaculation, such as retrograde ejaculation or erectile dysfunction.
iii. **Obstruction:** Blockages in the male reproductive tract, such as vas deferens absence or scarring.
iv. **Lifestyle Factors:** Factors like excessive alcohol consumption, smoking, drug use, obesity, exposure to toxins, or prolonged exposure to high temperatures can contribute to male infertility.

CAUSES OF FEMALE INFERTILITY:

i. **Ovulation Disorders:** Irregular or absent ovulation due to hormonal imbalances, PCOS, thyroid disorders, or premature ovarian failure.
ii. **Fallopian Tube Blockage:** Blockages or damage to the fallopian tubes due to conditions like pelvic inflammatory disease, endometriosis, or previous surgeries.
iii. **Uterine or Cervical Issues:** Abnormalities in the uterus or cervix, such as fibroids, polyps, or cervical stenosis, can affect fertility.
iv. **Endometriosis:** When uterine tissue grows outside the uterus, it can lead to inflammation, scarring, and structural abnormalities.
v. **Age-related Factors:** Advanced maternal age is associated with a decline in egg quality and quantity, increasing the risk of infertility and pregnancy complications.

These are some of the common causes of infertility in both males and females, but it's essential to consult with a healthcare provider for a thorough evaluation and diagnosis if infertility is a concern

vii) Differentiate between the following: (Tabular form)
(a) Testes and Ovaries (b) Spermatogenesis and Oogenesis

ANS: DIFFERENTIATE BETWEEN:

(a) TESTES VS. OVARIES

Aspect	Testes	Ovaries
LOCATION	In the scrotum (external)	In the pelvic cavity (internal)
FUNCTION	Produce sperm and testosterone	Produce eggs (ova) and female hormones
REPRODUCTIVE ROLE	Male reproductive organs	Female reproductive organs
GAMETES PRODUCED	Sperm	Eggs (ova)
HORMONES PRODUCED	Testosterone	Estrogen and progesterone
DEVELOPMENT AT BIRTH	Present, but non-functional	Present, but non-functional
ROLE IN PREGNANCY	No direct role in pregnancy	Essential for pregnancy

(b) SPERMATOGENESIS VS. OOGENESIS

ASPECT	SPERMATOGENESIS	OOGENESIS
LOCATION	Occurs in the testes	Occurs in the ovaries
GAMETES PRODUCED	Sperm	Eggs (ova)
NUMBER OF GAMETES	Four functional sperm per meiosis	One functional egg per meiosis
TIMING	Begins at puberty and continues throughout life	Begins before birth, pauses, and completes during menstrual cycles
FREQUENCY	Continuous and ongoing	Periodic, with long pauses
GENETIC VARIATION	High due to frequent divisions	Low due to limited divisions
ROLE IN REPRODUCTION	Contributes to fertilization	Provides eggs for fertilization

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

CHAPTER: 21 **DEVELOPMENT AND AGING**

ENCIRCLE THE CORRECT CHOICE:

i) The genetic material of a fertilized egg is a combination of
(a) The mother's DNA only (b) The father's DNA only
(c) Both the mother's and father's DNA (d) None of the above

ii) Human embryonic development occurs in the:
(a) Uterus (b) Fallopian tube (c) Cervix (d) Vagina

iii)The process of cell differentiation involves:
(a) The formation of gametes (b) The formation of zygotes
(c) The specialization of cells for different functions (d) The replication of DNA

iv) The implantation of the embryo in the uterus occurs at
(a) 1 week after fertilization (b) 2 weeks after fertilization
(c) 4 weeks after fertilization (d) 8 weeks after fertilization

v) Aging is characterized by:
(a) The accumulation of genetic mutations (b) The breakdown of cellular processes
(c) The cessation of mitosis (d) The reduction in DNA replication

vi) The inner cell mass of a blastocyst gives rise to:
(a) The placenta (b) The embryo (c) The umbilical cord (d) The amniotic fluid

vii) The process of fertilization occurs in the
(a) Uterus (b) Fallopian tube (c) Cervix (d) Vagina Answer

viii) The stages of embryonic development are:
(a) Cleavage, implantation, gastrulation, and organogenesis
(b) Fertilization, implantation, gastrulation, and organogenesis
(c) Cleavage, fertilization, gastrulation, and organogenesis
(d) Cleavage, fertilization, implantation, and organogenesis

ix) The placenta is responsible for:
(a) Providing oxygen and nutrients to the developing embryo
b) Removing waste products from the developing embryo
(c) Producing hormones that maintain the pregnancy
(d) All of the above

X) The formation of the neural tube occurs during
(a) Cleavage (b) Gastrulation (c) Organogenesis (d) Implantation

Q. NO:	ANSWER	Q. NO:	ANSWER
i)	(c) Both the mother's and father's DNA	vi)	(b) The embryo
ii)	(a) Uterus	vii)	(b) Fallopian tube
iii)	(c) The specialization of cells for different functions	viii)	(a) Cleavage, implantation, gastrulation, and organogenesis
iv)	(b) 2 weeks after fertilization	ix)	(d) All of the above
v)	(a) The accumulation of genetic mutations	x)	(b) Gastrulation

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

WRITE SHORT ANSWER OF THE FOLLOWING:

i) Why cleavage in birds and reptiles called meroblastic?

ANS: Cleavage in birds and reptiles is called meroblastic because it is an incomplete or partial type of cleavage. In these animals, the cleavage furrows do not penetrate the entire egg cell. Instead, cleavage occurs only in the cytoplasmic region near the animal pole of the egg, leaving a large yolk-filled portion called the vegetal pole relatively undivided. This incomplete cleavage is in contrast to holoblastic cleavage, where the entire egg undergoes complete cleavage into smaller blastomeres.

ii) Why morula is named so?

ANS: Morula is named so because it resembles a mulberry in terms of its appearance. The term "morula" is derived from the Latin word "morum," which means mulberry. Morula is an early stage in embryonic development where the fertilized egg undergoes multiple rounds of cell division, resulting in a compact cluster of cells with a spherical appearance resembling a mulberry.

iii) How embryonic tissues influence other embryonic tissues?

ANS: Embryonic tissues influence each other through a process known as induction. During development, certain groups of cells or tissues release signaling molecules or chemical cues that influence the development and differentiation of neighboring tissues. These signaling molecules can affect the fate, function, and differentiation of adjacent embryonic tissues, leading to the formation of complex structures and organ systems.

iv) Define organizers and differentiate between primary and secondary induction.

ANS: ORGANIZERS:

Organizers are specialized groups of cells or tissues that play a crucial role in embryonic development by influencing the fate and development of adjacent tissues. They secrete signaling molecules or chemical cues that affect the differentiation and development of neighboring cells and tissues, helping to orchestrate the formation of complex structures and organ systems during embryogenesis.

DIFFERENTIATE BETWEEN PRIMARY AND SECONDARY INDUCTION

ASPECT	ORGANIZERS	PRIMARY INDUCTION	SECONDARY INDUCTION
DEFINITION	Groups of cells or tissues that play a crucial role in embryonic development by influencing the fate and development of adjacent tissues.	The initial influence of organizers on nearby tissues, typically in early embryonic stages.	The cascading effects of primary induction, involving tissues that have already undergone primary induction on other tissues, often leading to further differentiation and development.
TIMING IN DEVELOPMENT	Present throughout embryonic development.	Occurs early in embryonic development.	Occurs later in embryonic development, following primary induction.
SIGNALING MOLECULES	Release signaling molecules or chemical cues that influence neighboring tissues.	Release signaling molecules to initiate cell differentiation and development in adjacent tissues.	Respond to signals from tissues that have undergone primary induction, transmitting further instructions for differentiation.
EXAMPLES	The Spemann organizer in amphibians plays a role in neural induction. The apical ectodermal ridge (AER) in limb development influences limb formation.	Spemann's organizer induces the formation of the neural tube in early amphibian embryos. AER induces the outgrowth and patterning of limb structures.	After the neural tube is induced, further signals from neural tissue guide the development of specific regions of the brain. After limb outgrowth is initiated, signals from developing limb tissues regulate the formation of specific limb structures.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

v) Differentiate between blastula and gastrula.

ANS: DIFFERENTIATE BETWEEN BLASTULA AND GASTRULA.

ASPECT	BLASTULA	GASTRULA
DEVELOPMENTAL STAGE	Early stage of embryonic development.	Follows the blastula stage in embryonic development.
FORMATION	Forms after several rounds of cleavage divisions of the fertilized egg.	Forms through a process called gastrulation, which transforms the blastula into a more complex structure.
OVERALL STRUCTURE	Hollow ball of cells with a central fluid-filled cavity (blastocoel).	Three primary germ layers develop: ectoderm, mesoderm, and endoderm.
CELL DIFFERENTIATION	Cells are undifferentiated and relatively uniform in structure.	Cells begin to differentiate into different cell types, marking the beginning of tissue and organ formation.
KEY EVENTS	Mainly a sphere with a blastocoel.	Formation of three germ layers: - Ectoderm: Outer layer - Mesoderm: Middle layer - Endoderm: Innermost layer
IMPORTANCE	Serves as an early stage where the embryo undergoes initial cleavage divisions.	Marks the transition to a more complex stage of development where tissue layers are established, laying the foundation for organ and body system formation.
EXAMPLE	In mammals, it is known as the blastocyst.	In non-mammalian vertebrates and many invertebrates, it is called the gastrula.

vi) Why monozygotic twins having same sex?

ANS: Monozygotic twins, also known as identical twins, are the result of a single fertilized egg (zygote) splitting into two embryos. These twins have the same genetic material because they originate from the same zygote. As a result, they are of the same sex because the sex of an individual is determined by the combination of sex chromosomes inherited from the parents. In monozygotic twins, the splitting of the zygote occurs shortly after fertilization, before any sex determination has taken place, so both embryos will have the same sex chromosome configuration, whether it's two X chromosomes (female) or one X and one Y chromosome (male).

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

CHAPTER: 22 **CHROMOSOME AND DNA**

ENCIRCLE THE CORRECT ANSWER:

i) W. Sutton and Theodor observed this behavior of chromosomes.
(a) Formation (b) Distribution (c) Staining (d) Replication

ii) Which is not the part of chromosomal theory of inheritance.
(a) Gametes do not make equal hereditary contribution (b) Chromosome segregate during meiosis
(c) Nucleus is the room of hereditary material (d) Gametes have one copy of homologous chromosome

iii) Newly formed chromosome has two.
(a) Chromatids (b) Heterologous (c) Centromere (d) Pairs

iv) The complex of histone octamer and two loops of duplex DNA is.
(a) Chromomer (b) Chromatin (c) Nucleosome (d) Heterochromatin

v) Small part of DNA which has information to synthesize specific polypeptide chain is.
(a) Genome (b) Locus (c) Gene (d) Nucleotide

vi) According to replication model, the parental double helix remains intact and conserved called.
(a) Semi conservative (b) Conservative (c) Disposed (d) Eukaryotic replication

vii) A protein which prevents the re-binding of complementary strand during replication at fork is.
(a) DNA helicase (b) Ligase (c) SSB (d) Primase

viii) The DNA strand which continuously replicate in the direction of 5'-3' called.
(a) Lagging strand (b) Leading strand (c) Primer (d) Okazaki fragment

ix) Process where information present on gene is used to produce a functional product by living organism called.
(a) Transcription (b) Translation (c) Gene expression (d) Regulation of gene expression

x) The set of rules used to store the genetic information within a DNA for particular protein synthesis is.
(a) Gene (b) Genetic codes (c) Codon (d) Anticodon

Q. no:	Answer	Q. no:	Answer
i)	(b) Distribution	vi)	(a) Semi conservative
ii)	(c) Nucleus is the room of hereditary material	vii)	(c) SSB
iii)	(a) Chromatids	viii)	(b) Leading strand
iv)	(c) Nucleosome	ix)	(c) Gene expression
v)	(c) Gene	x)	(b) Genetic codes

WRITE SHORT ANSWER OF THE FOLLOWING:

i) Why DNA is negatively charged molecules?
ANS: DNA is negatively charged because it contains phosphate groups in its backbone. Phosphate groups are acidic and carry a negative charge due to the presence of negatively charged oxygen atoms. This gives the DNA molecule an overall negative charge.

ii) Why replication is called semi-conservative process?
ANS: Replication is called a semi-conservative process because, during DNA replication, each newly synthesized DNA molecule consists of one original (parental) strand and one newly synthesized (daughter) strand. The original strand serves as a template for the synthesis of the new strand. This ensures that the genetic information is conserved from

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

one generation of DNA molecules to the next, with half of the original genetic material being preserved in each daughter molecule.

iii) Do genes of prokaryotic cell and eukaryotic cells are different in structure? If so give the main differences between them?

ANS: Yes, genes in prokaryotic and eukaryotic cells have structural differences. In prokaryotic cells, genes are typically organized as continuous stretches of DNA that code for proteins without intervening non-coding sequences (introns). In contrast, eukaryotic genes often contain introns, which are non-coding sequences interspersed among coding sequences (exons). Eukaryotic genes are also associated with regulatory elements and are more complex in structure compared to prokaryotic genes.

iv) What are leading and lagging strand of DNA?

ANS: The leading strand and lagging strand are two strands of DNA being replicated during DNA replication. The leading strand is synthesized continuously in the 5' to 3' direction because it is oriented in the direction of DNA unwinding. The lagging strand is synthesized discontinuously in short fragments called Okazaki fragments, which are later joined together. This occurs because the lagging strand is oriented in the opposite direction of DNA unwinding.

v) Give the name of enzymes involved in replication of DNA, also give their brief functions.

ANS: Enzymes involved in DNA replication include DNA polymerase, primase, helicase, and DNA ligase.

- **DNA polymerase:** Synthesizes the new DNA strand by adding complementary nucleotides to the template strand.
- **Primase:** Synthesizes a short RNA primer that serves as a starting point for DNA synthesis.
- **Helicase:** Unwinds the DNA double helix.
- **DNA ligase:** Joins the Okazaki fragments on the lagging strand and seals nicks in the DNA backbone.

vi) Do introns are transcribed, and involve in translation?

ANS: Introns are transcribed but are typically removed during a process called splicing before the mRNA is translated. Introns do not directly participate in translation, as they are non-coding regions of the mRNA.

vii) Give changes occur in mRNA during transport from nucleus to cytoplasm.

ANS: During transport from the nucleus to the cytoplasm, mRNA undergoes post-transcriptional modifications, including the addition of a 5' cap and a 3' poly-A tail. These modifications protect the mRNA from degradation, facilitate its export from the nucleus, and aid in translation initiation in the cytoplasm.

viii) Give site at ribosome and their functions during translation.

ANS: The ribosome consists of two subunits, the small subunit (40S in eukaryotes) and the large subunit (60S in eukaryotes). During translation, the small subunit binds to the mRNA, while the large subunit catalyzes peptide bond formation between amino acids. The ribosome has several binding sites:

- **A site (aminoacyl-tRNA site):** Binds incoming aminoacyl-tRNA molecules.
- **P site (peptidyl-tRNA site):** Holds the tRNA carrying the growing polypeptide chain.
- **E site (exit site):** Allows for the exit of the deacylated tRNA.

ix) Draw structure of tRNA and give functions of their different sites.

ANS: STRUCTURE OF tRNA:

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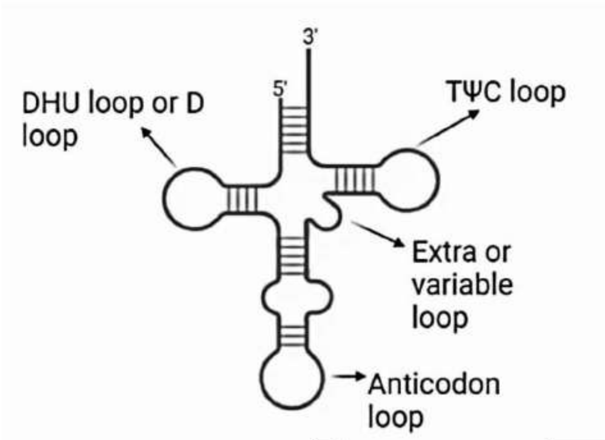

FUNCTIONS OF DIFFERENT SITES ON TRNA:

i. **Acceptor (A) Site:** The A site of tRNA is responsible for binding to the amino acid, forming an aminoacyl-tRNA complex. It plays a crucial role in bringing the appropriate amino acid to the ribosome during protein synthesis.

ii. **Peptidyl (P) Site:** The P site of tRNA holds the growing polypeptide chain. It ensures that the amino acids are correctly linked together to form a protein by facilitating peptide bond formation between the incoming amino acid and the growing chain.

iii. **Exit (E) Site:** The E site of tRNA is where the discharged tRNA exits the ribosome after it has donated its amino acid to the growing polypeptide chain. This site prepares the tRNA for its return to the cytoplasm to pick up another amino acid.

These distinct tRNA sites are essential for the precise and efficient synthesis of proteins during translation.

X) Do mutation is always harmful? Justify you answer.

ANS: Mutations are not always harmful. Some mutations may have neutral or even beneficial effects. The impact of a mutation depends on its location in the genome and the specific changes it introduces. Harmful mutations can lead to diseases or disorders, while neutral mutations may have no discernible effect on an organism's phenotype. Beneficial mutations can provide an advantage in certain environments and contribute to evolution.

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CHAPTER: 23 **INHERITANCE**

CIRCLE THE CORRECT ANSWER:

i) Which of the following would cause phenotypic variations among organisms of the same genotype?

(a) Continuous variation within the species same species (b) Different varieties of the
(c) Different sexes (d) Exposure to different environments

ii) In which of the following examples of human inheritance is the inheritance pattern explained by multiple alleles on an autosomal chromosome?

(a) The ABO blood group system (b) Cystic fibrosis (c) Down's syndrome (d) Haemophilia

iii) What are the phenotypes of the parents of a colour-blind son and a non-carrier daughter with normal colour vision?

	Father	Mother
(a)	Carrier	Normal
(b)	Colour-blind	Carrier
(c)	Colour-blind	Colour-blind
(d)	Normal	Carrier

iv) In the F2 generation of a dihybrid cross, the phenotypes occurred in the ratio 3.1 what does this result indicate?

(a) The alleles were segregating independently (b) Polygenic inheritance was involved
(c) Codominance was being shown (d) The gene loci were linked

v) Two parents, both of blood group A. have a daughter of blood group O. What is the probability that their next child will be a boy who has blood group O?

(a) 0.25 (b) 0.5 (c) 0.75 (d) 1

vi) A boy is colour blind which could be the genotype of his mother?

(a) X^NX^N (b) X^NX^n (c) X^NY (d) X^nY

vii) Number of chromosome in grass hopper female is 24. How many chromosomes are present in grass hopper male?

(a) 26 (b) 25 (c) 24 (d) 23

viii) The allele of holandric gene is located at?

(a) X-Chromosome (b) Y-Chromosome (c) Autosome (d) None of these

ix) The gene which interferes and masks the phenotype of the phenotype of the non-locus gene called?

(a) Mutant gene (b) Epistatic gene (c) Pleiotropy (d) Sex-linked

x) International society of blood transfusion has found more than?

(a) 10 blood system (b) 20 blood system (c) 30 blood system (d) 40 blood system

Q. no:	Answer	Q. no:	Answer
i)	(d) Exposure to different environments	**vi)**	(b) X^NX^n
ii)	(a) The ABO blood group system	**vii)**	(b) 25
iii)	(b) Colour-blind Carrier	**viii)**	(b) Y-Chromosome
iv)	(a) The alleles were segregating independently	**ix)**	(b) Epistatic gene
v)	(c) 0.75	**x)**	(c) 30 blood system

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

WRITE SHORT ANSWER OF THE FOLLOWING:

i) Why Rh-incompatibility could be a danger to the developing foetus and mother?

ANS: Rh-incompatibility can be dangerous to the developing fetus and mother because it involves a mismatch between the Rh factor in the mother's and fetus's blood. If the mother is Rh-negative and the fetus is Rh-positive, the mother's immune system may produce antibodies against the fetal blood cells, potentially causing severe complications for the fetus, such as hemolytic anemia and jaundice. This condition, known as hemolytic disease of the newborn, can be life-threatening for the baby and poses risks in subsequent pregnancies as well.

ii) Why haemophilia is common in human male?

ANS: Hemophilia is common in human males because it is a sex-linked recessive genetic disorder. The genes responsible for hemophilia are located on the X chromosome. Males have only one X chromosome (XY), so if they inherit a defective X chromosome carrying the hemophilia gene, they will develop the disorder. Females, with two X chromosomes (XX), are less likely to have hemophilia because they can carry a healthy gene on one X chromosome, often compensating for the defective gene on the other.

iii) What do you mean by sex-influenced trait?

ANS: A sex-influenced trait is a genetic characteristic that is influenced by an individual's sex but is not limited to one sex. These traits are affected by sex hormones and can result in different expressions in males and females. For example, male pattern baldness is a sex-influenced trait; it is influenced by hormones, with males being more prone to baldness due to the effects of male sex hormones. However, females can also experience hair loss but typically in a different pattern due to their hormonal differences.

iv) Differentiate between following:

a) Linkage and crossing over

b) Monohybrid Cross and Dihybrid Cross

c) X-linked trait and Y-linked trait

d) Autosomes and sex-chromosome

e) Incomplete dominance and Co-dominance

ANS: DIFFERENTIATE BETWEEN:

a) LINKAGE AND CROSSING OVER

ASPECT	LINKAGE	CROSSING OVER
DEFINITION	Genes located close on the same chromosome tend to be inherited together.	Exchange of genetic material between homologous chromosomes during meiosis.
OCCURRENCE	Common in genes located near each other on the same chromosome.	Occurs between homologous chromosomes during prophase I of meiosis.
EFFECT ON ALLELES	Alleles in linked genes remain together, and new combinations are rare.	Alleles in crossing-over can lead to the creation of new allele combinations.
GENETIC MAPPING	Less useful for genetic mapping as linked genes do not assort independently.	Useful for genetic mapping as it helps determine the relative distances between genes on a chromosome.

b) MONOHYBRID CROSS AND DIHYBRID CROSS

ASPECT	MONOHYBRID CROSS	DIHYBRID CROSS
NUMBER OF TRAITS STUDIED	Involves the study of a single trait with two alleles.	Involves the study of two different traits, each with two alleles.
PUNNETT SQUARE	Requires a 2x2 Punnett square.	Requires a 4x4 Punnett square.
PHENOTYPIC RATIOS	Typically results in a 3:1 phenotypic ratio for the trait being studied.	Results in a 9:3:3:1 phenotypic ratio for both traits being studied.
EXAMPLE	Cross between two individuals with different flower color alleles (e.g., Aa x Aa).	Cross between two individuals with different flower color and plant height alleles (e.g., AaBb x AaBb).

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c) X-LINKED TRAIT AND Y-LINKED TRAIT

ASPECT	X-LINKED TRAIT	Y-LINKED TRAIT
CHROMOSOME	Located on the X chromosome.	Located on the Y chromosome.
INHERITANCE PATTERN	Can be passed from mother to both sons and daughters, but not from father to sons.	Passed only from father to sons.
EXAMPLES	Color blindness and hemophilia in humans.	Some male-specific traits in animals, such as certain sex-determining genes in fruit flies.

d) AUTOSOMES AND SEX CHROMOSOMES

ASPECT	AUTOSOMES	SEX CHROMOSOMES
Chromosome Type	Non-sex chromosomes, present in both sexes.	Determine an individual's sex (X and Y chromosomes in humans).
Inheritance	Inherited equally from both parents.	Inherited unequally; sex chromosomes are inherited differently based on the sex of the parent.
Number	In humans, there are 44 autosomes (22 pairs).	In humans, there are 2 sex chromosomes (X and Y).
Function	Carry genes responsible for most body traits.	Determine an individual's biological sex and carry genes related to sex determination and some sex-linked traits.

e) INCOMPLETE DOMINANCE AND CO-DOMINANCE

ASPECT	INCOMPLETE DOMINANCE	CO-DOMINANCE
ALLELE INTERACTION	Neither allele is completely dominant over the other; heterozygotes show an intermediate phenotype.	Both alleles are equally dominant, and heterozygotes exhibit the expression of both alleles simultaneously.
PHENOTYPIC RATIOS	Typically results in a 1:2:1 phenotypic ratio in the F2 generation.	Results in a 1:2:1 phenotypic ratio in the F2 generation as well.
EXAMPLE	Red and white flower alleles produce pink flowers in snapdragons (RW x RW).	ABO blood group system in humans, where both A and B alleles are co-dominant (I^AI^B results in AB blood type).

v) Why human male are heterogametic?

ANS: In humans, males are heterogametic, meaning they have two different sex chromosomes, X and Y (XY). This heterogametic condition arises during fertilization when the male's sperm can carry either an X or a Y chromosome. Females, on the other hand, are homogametic, as they have two X chromosomes (XX). This difference in sex chromosome composition determines the sex of the offspring. The presence of a Y chromosome leads to the development of a male, while the absence of a Y chromosome results in a female. This heterogametic sex determination system is common in many species, including mammals, and it contrasts with the homogametic system found in species where females have identical sex chromosomes.

vi) What do you mean by dichromacy and monochromacy?

ANS: Dichromacy and monochromacy are two types of color vision deficiencies:

i. **DICHROMACY:** Dichromats are individuals who have only two types of functioning color receptors or cones in their eyes instead of the usual three. As a result, they have difficulty distinguishing between certain colors. The most common form of dichromacy is red-green color blindness, where the individual cannot differentiate between red and green hues. Another form is blue-yellow color blindness. Dichromacy is a partial color vision deficiency.

ii. **MONOCHROMACY:** Monochromats have only one type of functioning cone or no functioning cones at all in their eyes. They perceive the world in shades of gray and are completely colorblind. Monochromacy is a more severe form of color vision deficiency compared to dichromacy.

vii) What do you known about ZZ and ZW in sex determination?

ANS: ZZ and ZW are sex determination systems found in certain animals, particularly in birds, reptiles, and some insects. They are the opposite of the more common XY and XX sex determination system found in mammals, including humans.

In the ZZ/ZW system:

- Males have two similar sex chromosomes, usually denoted as ZZ.
- Females have two different sex chromosomes, usually denoted as ZW.

In this system, it's the presence or absence of the Z chromosome that determines the sex of the offspring. If an individual inherits two Z chromosomes (ZZ), they develop as males. If they inherit one Z and one W chromosome (ZW), they develop as females. This is different from the XY/XX system where the presence of a Y chromosome determines male development.

viii) Why incomplete dominance is called partial dominance?

ANS: Incomplete dominance is called "partial dominance" because it describes a situation in genetics where neither of the two alleles in a heterozygous individual (one with two different alleles) is completely dominant over the other. Instead, both alleles exert some influence on the trait, resulting in an intermediate or blended phenotype. In incomplete dominance, the phenotype of the heterozygous individual falls between the phenotypes of the homozygous dominant and homozygous recessive individuals.

For example, in a flower color trait, if the dominant allele (R) produces red flowers and the recessive allele (r) produces white flowers, a heterozygous individual (Rr) may produce pink flowers. Neither allele is fully dominant over the other, hence the term "incomplete dominance" or "partial dominance." This contrasts with complete dominance, where one allele completely masks the expression of the other.

ix) Define following terms:

(a) Genetics	**(b) Allele**	**(c) Gene**
(d) Karyotype	**(e) multiple allele**	**(f) test cross**
(g) Mutant	**(h) Erythroblastosis foetilus**	**(i) Back cross**
(j) Linkage	**(k) Crossing over**	**(l) Protanopi**
(m) Isoagglutinogen		

ANS: **DEFINITIONS OF THE TERMS:**

(a) Genetics: Genetics is the scientific study of heredity, genes, and the variation of traits in living organisms. It explores how traits are passed from one generation to the next and how genetic information is inherited and expressed.

(b) Allele: An allele is one of the variant forms of a gene, located at a specific position on a chromosome. Alleles can differ in their DNA sequence and can lead to variations in traits or characteristics.

(c) Gene: A gene is a segment of DNA that contains the instructions for building a specific protein or performing a particular function in an organism. Genes are the basic units of heredity and determine an organism's traits.

(d) Karyotype: A karyotype is a visual representation of an individual's chromosomes, organized by size, shape, and other characteristics. It is used to examine the number and structure of chromosomes in an organism, often for diagnostic purposes.

(e) Multiple Allele: Multiple alleles refer to the existence of more than two different alleles for a specific gene in a population. However, each individual still carries only two alleles—one from each parent—for a given gene.

(f) Test Cross: A test cross is a genetic cross between an individual with a dominant phenotype but unknown genotype and an individual with a homozygous recessive genotype. It is used to determine the genotype of the first individual.

(g) Mutant: A mutant is an organism or genetic variant that carries a change or mutation in its DNA sequence compared to the typical or wild-type form. Mutations can lead to altered traits or characteristics.

(h) Erythroblastosis Fetalis: Erythroblastosis fetalis is a condition in which a pregnant woman's immune system produces antibodies against the red blood cells of her developing fetus. This can occur when there is a blood group incompatibility between the mother and the fetus, often related to the Rh factor.

(i) Back Cross: A back cross is a genetic cross between an individual of interest and one of its parent or a genetically similar organism. It is used to introduce or maintain specific traits or genes in a breeding population.

(j) Linkage: Linkage refers to the tendency of certain genes located on the same chromosome to be inherited together more frequently than expected by random chance. This is because they are physically close to each other on the chromosome.

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(k) Crossing Over: Crossing over is a genetic process that occurs during meiosis when homologous chromosomes exchange segments of genetic material. This leads to genetic recombination and increased genetic diversity in offspring.

(l) Protanopia: Protanopia is a form of color blindness in which individuals have difficulty distinguishing between red and green colors due to a defect in the cones of the eye that detect these colors.

(m) Isoagglutinogen: Isoagglutinogens are antigens found on the surface of red blood cells that can trigger an immune response if they are not compatible with the recipient's blood type. They are an important factor in blood transfusion compatibility.

x) Why law of segregation is called purity of gametes?

ANS: THE LAW OF SEGREGATION

The Law of Segregation, proposed by Gregor Mendel, states that during the formation of gametes (sperm and egg cells), the two alleles of a gene segregate or separate from each other so that each gamete carries only one allele for that gene. This separation ensures that each gamete is "pure" in the sense that it contains only one version of the gene.

The term "purity of gametes" refers to the fact that each gamete produced by an organism carries a single, unaltered allele for each gene. This is significant because when these gametes combine during fertilization, they can recombine in various combinations, leading to genetic diversity in the offspring. Without the segregation of alleles, genetic information would not be properly mixed and passed on to the next generation, which would result in a lack of genetic diversity and ultimately hinder the process of evolution.

CHAPTER: 24 **EVOLUTION**

ENCIRCLE THE MOST APPROPRIATE RESPONSE.

i) Which of the following was supporting the theory of special creation?
(a) George Buffon (b) Lamarck (c) Darwin (d) Suárez

ii) Archaeopteryx is a connecting link between
(a) Amphibia and reptiles (b) Reptiles and Aves (c) Aves and mammals (d) Fish and amphibia

iii) All of the following are homologous organs except:
(a) Wings of bat (b) Wings of butterfly (c) Wings of bird (d) Flippers of dolphin

iv) Which of the following are analogous organs except:
(a) Wings of insect (b) Wings of bird (c) Wings of bat (d) Arm of man

v) Life originated in
(a) Air (b) Water (c) Land (d) Space

vi) Philosophie Zoologique was written by
(a) Wallace (b) Darwin (c) Weismann (d) Lamarck

vii) Which of the following is incorrect regarding Lamarck's theory of Inheritance of acquired characters?
(a) Use and disuse of the organs inherited
(b) Continuously using some organ results in its further strengthening in offspring
(c) Continuously disuse of some organ results in its weakening in offspring
(d) Concrete support to the theory through experiments

viii) The theory on human population which inspired Darwin was proposed by:
(a) Malthus (b) Lyell (c) Hutton (d) Wallace

ix) Theory of Natural Selection was lacking any support from:
(a) Biogeography (b) Genetics (c) Comparative anatomy (d) Molecular Biology

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x) Which of the following cannot be the disruptive force for Hardy-Weinberg Theorem?

(a) Selection (b) Non-random mating (c) Mutation (d) Random mating

Q. no:	Answer	Q. no:	Answer
i)	(d) Suárez	vi)	(d) Lamarck
ii)	(b) Reptiles and Aves	vii)	(d) Concrete support to the theory through experiments
iii)	(b) Wings of butterfly	viii)	(a) Malthus
iv)	(d) Arm of man	ix)	(d) Molecular Biology
v)	(b) Water	x)	(d) Random mating

WRITE SHORT ANSWERS OF THE FOLLOWING QUESTIONS.

i) Differentiate between the following:

a) Special Creation and Evolution

b) Lamarckism and Darwinism

c) Invagination theory and Endosymbiotic theory

d) Bottle-neck effect and Founder's effect

e) Convergent and Divergent evolution

ANS: DIFFERENTIATE BETWEEN

a) SPECIAL CREATION VS. EVOLUTION:

ASPECT	SPECIAL CREATION	EVOLUTION
EXPLANATION OF ORIGIN	Divine creation by a deity(s)	Gradual change over time
MECHANISM	Supernatural intervention	Natural selection, mutation, etc.
TIME SCALE	Typically, a short and fixed time	Takes place over millions of years
EVIDENCE	Based on religious texts and faith	Supported by fossil and genetic evidence
MAIN FIGURES	Creationists, theologians	Charles Darwin, Alfred Wallace, modern biologists

b) LAMARCKISM VS. DARWINISM:

ASPECT	LAMARCKISM	DARWINISM
INHERITANCE OF TRAITS	Acquired traits can be inherited	Inherited traits determine fitness
MECHANISM	Use and disuse of organs, inheritance of acquired characteristics	Natural selection based on variation
TIME PERIOD	Proposed in the early 19th century	Developed by Charles Darwin in the mid-19th century
ACCEPTANCE	Widely discredited in modern biology	Widely accepted in modern biology

c) INVAGINATION THEORY VS. ENDOSYMBIOTIC THEORY:

ASPECT	INVAGINATION THEORY	ENDOSYMBIOTIC THEORY
MAIN IDEA	Suggests that early organisms formed through the folding in of cell membranes	Proposes that eukaryotic organelles, like mitochondria and chloroplasts, originated as free-living prokaryotic cells
EVIDENCE	Limited supporting evidence, not widely accepted	Supported by molecular and genetic evidence, widely accepted in biology
KEY FIGURES	Ernst Haeckel (proponent)	Lynn Margulis (proponent)

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d) BOTTLENECK EFFECT VS. FOUNDER'S EFFECT:

Aspect	Bottleneck Effect	Founder's Effect
OCCURS WHEN	Population drastically reduced in size	A small group of individuals establishes a new population
GENETIC DIVERSITY	Results in loss of genetic diversity	Can lead to reduced genetic diversity in the new population
CAUSE	Typically due to a catastrophic event	Often occurs during colonization or migration
CONSEQUENCES	Increases the risk of genetic diseases	Can lead to genetic drift and unique allele frequencies

e) CONVERGENT EVOLUTION VS. DIVERGENT EVOLUTION:

ASPECT	CONVERGENT EVOLUTION	DIVERGENT EVOLUTION
DEFINITION	Unrelated species develop similar traits in response to similar environmental pressures	A single species or ancestral population evolves into different species
RESULTING TRAITS	Similarities in traits or features in different species	Differences in traits or features among related species
EXAMPLES	Wings of birds, bats, and insects for flying	Darwin's finches with different beak shapes for varied diets
RELATIONSHIP	Species are not closely related on the phylogenetic tree	Species share a common ancestor and then diverge

ii) How the homologous organs support the theory of evolution?

ANS: Homologous organs support the theory of evolution because they are structures found in different species that share a common ancestry. These organs have similar anatomical features and often perform similar functions, but they may have adapted differently in response to the specific needs of each species. The presence of homologous organs suggests that species have evolved from a common ancestor, and as they diversified over time, these organs were modified to suit the ecological niches of each species. For example, the forelimbs of vertebrates, such as humans, bats, whales, and cats, have similar bone structures, indicating a common evolutionary origin despite differences in function (e.g., flying, swimming, walking).

iii) What are analogous organs? Give example.

ANS: Analogous organs are structures that have similar functions in different species but have different evolutionary origins. These organs are not evidence of a common ancestor but instead result from convergent evolution, where unrelated species independently evolve similar traits due to similar environmental pressures. An example of analogous organs is the wings of birds and the wings of insects. While both allow for flying, they have different anatomical structures and evolved separately.

iv) How Alfred Wallace contributed Charles Darwin regarding the Natural Selection?

ANS: Alfred Wallace independently developed the theory of natural selection, and his contributions played a significant role in shaping Charles Darwin's thinking about evolution. Wallace and Darwin arrived at similar conclusions about the process of evolution through natural selection independently. In 1858, both scientists jointly presented their ideas to the Linnean Society of London. Wallace's work, including his observations of biogeography and the distribution of species, helped confirm and strengthen the concept of natural selection. Darwin and Wallace's joint presentation is often seen as a pivotal moment in the development of evolutionary theory.

v) What do understand by the "descent with modification"?

ANS: "Descent with modification" is a fundamental concept in the theory of evolution. It refers to the idea that species have evolved from common ancestors and have undergone changes or modifications over time. As new generations of organisms are produced, they may inherit variations or traits from their ancestors, but these traits can be modified or

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adapted to suit the specific environmental conditions and challenges they face. Over long periods, these modifications accumulate, leading to the diversity of life on Earth.

vi) How Neo-Darwinism differs from Darwinism?

ANS: Neo-Darwinism, also known as the modern synthesis, differs from Darwinism in that it incorporates genetics into the theory of evolution. While Charles Darwin proposed the idea of natural selection as the mechanism for evolution, he did not have access to the understanding of genetics as we do today. Neo-Darwinism combines Darwin's concept of natural selection with the principles of Mendelian genetics, providing a more comprehensive explanation of how traits are inherited and how variation arises within populations. It also incorporates the role of mutations as a source of genetic variation.

vii) How the sympatric speciation differs from parapatric speciation?

ANS: Sympatric speciation and parapatric speciation are two different modes of speciation:

Sympatric Speciation: In sympatric speciation, new species arise within the same geographic area or habitat. This can occur when a subset of individuals within a population develops distinct traits or behaviors that prevent them from interbreeding with the rest of the population. Factors like polyploidy (having multiple sets of chromosomes) or niche differentiation (occupying different ecological niches) can drive sympatric speciation.

Parapatric Speciation: Parapatric speciation occurs when new species evolve in adjacent but not entirely overlapping geographic areas. Gene flow between the adjacent populations is limited, often due to differences in ecological conditions or other barriers, leading to the development of distinct traits and eventually separate species.

viii) Justify Lamarck as an early proponent of evolution.

ANS: Lamarck is considered an early proponent of evolution because he proposed one of the earliest comprehensive theories of evolution in the early 19th century. Lamarck's ideas, known as Lamarckism or the inheritance of acquired characteristics, posited that organisms could change over their lifetimes in response to their environment, and these acquired traits could be passed on to their offspring. While his specific mechanism has been largely discredited, Lamarck's work laid the groundwork for later evolutionary thinking and inspired further research in the field of biology.

ix) What is genetic drift?

ANS: Genetic drift is a random process that can cause changes in allele frequencies within a population over generations. It occurs due to chance events rather than natural selection. Genetic drift is more pronounced in small populations and can lead to the loss of genetic diversity. There are two main forms of genetic drift:

Bottleneck Effect: This occurs when a population is significantly reduced in size due to a catastrophic event, such as a natural disaster or disease outbreak. The surviving individuals may have a limited subset of the original population's genetic diversity, leading to a loss of genetic variation.

Founder's Effect: This happens when a small group of individuals from a larger population establishes a new population in a different geographic area. The genetic makeup of the founder population may not represent the full genetic diversity of the original population, leading to differences in allele frequencies in the new population.

x) List out the factors affecting allele frequency

ANS: Factors affecting allele frequency in a population include:

i. **Natural Selection:** Differential survival and reproduction of individuals with certain traits can lead to changes in allele frequencies over time as advantageous alleles become more common.
ii. **Mutation:** New genetic variations can arise through mutations. These mutations can introduce new alleles into a population.
iii. **Gene Flow:** The movement of individuals or genes between different populations can change allele frequencies in both populations.
iv. **Genetic Drift:** Random changes in allele frequencies due to chance events, particularly in small populations.
v. **Non-Random Mating:** When individuals do not mate randomly within a population, it can lead to changes in allele frequencies.
vi. **Migration:** The movement of individuals between populations can introduce new alleles and change allele frequencies.

vii. **Environmental Factors:** Environmental changes can affect the fitness of certain traits, influencing which alleles are favored and thus affecting allele frequencies.

CHAPTER: 25 **MAN AND HIS ENVIRONMENT**

ENCIRCLE THE CORRECT CHOICE.

i) Which one of these mainly causes the greenhouse effect?
(a) Acid rain (b) Ozone layer depletion (c) Global warming (d) Forest fire

ii) Factors that affect the flow of energy at the trophic level are:
(a) Non-predatory deaths (b) Growth & Reproduction (c) Heat loss (d) Sunlight

iii) The main cause of the increase in the amount of CO_2 in the Earth's atmosphere is
(a) Rapidly growing human population (b) Burning of a large number of fossil fuels
(c) Increase in the number of industries (d) Decrease in the forested resources

iv) All is positive for which of the following processes?
(a) Oxidation of Nitrogen (b) Melting of ice and evaporation of water
(c) Oxidation of gold (d) Burning of chlorine

v) Which of the following dietary requirements is involved in nitrogen balance?
(a) Carbohydrates (b) Vitamins (c) Proteins (d) Essential fatty acids

vi) Which of the following cycle is not a gaseous type of cycle:
(a) Carbon cycle (b) Nitrogen cycle (c) Phosphorus cycle (d) Oxygen cycle

vii) The source of carbon to plants in the carbon cycle is
(a) Carbonate rocks (b) atmospheric carbon dioxide (c) Fossil fuels (d) all of the above

viii) Conversion of nitrates to nitrogen is called
(a) Nitrification (b) Denitrification (c) Ammonification (d) Nitrogen fixation

ix) what is the only source of energy for all ecosystems on Earth?
(a) Water (b) Sun (c) Animals (d) Plants

x) The pioneers in xerarch succession are the
(a) Foliose lichens (b) Mosses (c) Crustose lichens (d) Shrubs

Q. no:	Answer	Q. no:	Answer
i)	(b) Ozone layer depletion	vi)	(d) Oxygen cycle
ii)	(b) Growth & Reproduction	vii)	(b) atmospheric carbon dioxide
iii)	(b) Burning of a large number of fossil fuels	viii)	(b) Denitrification
iv)	(b) Melting of ice and evaporation of water	ix)	(b) Sun
v)	(c) Proteins	x)	(c) Crustose lichens

WRITE SHORT ANSWERS OF THE FOLLOWING QUESTIONS.

i) Why biogeochemical cycle is named so?

ANS: The term "biogeochemical cycle" is used because these cycles involve the movement and exchange of elements (chemicals) between the living organisms (bio), the Earth's surface (geo), and the atmosphere. These cycles describe the pathways that elements like carbon, nitrogen, phosphorus, and water take as they circulate through the biosphere, geosphere, and atmosphere, influencing the chemical composition of the Earth.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

ii) Differentiate between nitrification and de-nitrification?

ANS: DIFFERENTIATE BETWEEN NITRIFICATION AND DE-NITRIFICATION:

ASPECT	NITRIFICATION	DE-NITRIFICATION
PROCESS	Conversion of ammonia (NH3) or ammonium (NH4+) into nitrite (NO2-) and then nitrate (NO3-) by nitrifying bacteria (Nitrosomonas and Nitrobacter).	Conversion of nitrate (NO3-) into nitrogen gas (N2) or nitrous oxide (N2O) by denitrifying bacteria (e.g., Pseudomonas and Paracoccus).
OXYGEN REQUIREMENT	Aerobic (requires oxygen)	Anaerobic (occurs in the absence of oxygen)
ROLE	Part of the nitrogen cycle, converting ammonia produced by decomposition into forms usable by plants.	Returns nitrogen to the atmosphere, reducing the availability of nitrogen compounds in ecosystems.

iii) How does energy flow between trophic levels?

ANS: Energy flows through trophic levels in an ecosystem in a unidirectional manner. Producers, such as plants, capture solar energy through photosynthesis and convert it into chemical energy stored in organic compounds (e.g., glucose). Herbivores, primary consumers, feed on producers, obtaining energy by consuming these plants. Carnivores, secondary consumers, in turn, obtain energy by consuming herbivores, and so on. Energy is transferred as organisms are eaten and incorporated into the consumer's body. However, energy is lost at each trophic level due to metabolic processes, heat production, and waste, resulting in a pyramid-like structure of energy distribution.

iv) What are the major producers in a terrestrial ecosystem?

ANS: The major producers in a terrestrial ecosystem are plants. These can include various types of vegetation, such as trees, shrubs, grasses, and other photosynthetic organisms. Plants are capable of photosynthesis, which allows them to convert sunlight, water, and carbon dioxide into organic compounds (mainly glucose) that serve as the primary source of energy for the entire ecosystem.

v) How does ecological succession affect the community?

ANS: Ecological succession is the process by which a community of organisms gradually changes over time in response to environmental disturbances or changes. It can have several effects on the community:

i. **Increased Diversity:** Over time, succession often leads to increased species diversity as different species become adapted to various stages of the changing environment.
ii. **Changes in Species Composition:** The types of species present in the community can change as new species colonize and others are replaced. Pioneer species are typically replaced by more competitive and mature species.
iii. **Changes in Ecosystem Structure:** As succession progresses, there can be changes in physical structures of the ecosystem, such as soil development, plant height, and canopy cover.
iv. **Increased Stability:** Mature, climax communities tend to be more stable and resilient to disturbances compared to early successional stages.

vi) What are the causes of ozone layer depletion?

ANS: Ozone layer depletion is primarily caused by the release of human-made chemicals known as ozone-depleting substances (ODS). The main causes include:

i. **Chlorofluorocarbons (CFCs):** These synthetic compounds, used in refrigerants, aerosol propellants, and foam-blowing agents, release chlorine atoms when they break down in the stratosphere. Chlorine atoms catalytically destroy ozone molecules.
ii. **Halons:** Halons are similar to CFCs and are used in fire extinguishers. They release bromine and chlorine when they reach the stratosphere, leading to ozone depletion.
iii. **Carbon Tetrachloride and Methyl Chloroform:** These industrial chemicals also release chlorine and contribute to ozone layer depletion.
iv. **Nitrous Oxide:** Nitrous oxide emissions from industrial processes and agriculture can indirectly affect the ozone layer by releasing nitrogen oxides, which participate in ozone destruction.

These chemicals break down ozone molecules in the stratosphere, leading to the thinning of the ozone layer and the formation of the ozone hole.

vii) Why should we conserve biodiversity?

ANS: Biodiversity conservation is essential for several reasons:

i. **Ecosystem Stability:** Biodiversity contributes to the stability and resilience of ecosystems. Diverse ecosystems are better able to withstand environmental changes and disturbances.
ii. **Economic Value:** Many species provide direct economic benefits, such as food, medicine, and raw materials. Biodiversity also supports industries like agriculture and tourism.
iii. **Ecosystem Services:** Biodiversity is critical for the provision of ecosystem services, including pollination, nutrient cycling, water purification, and climate regulation.
iv. **Scientific Understanding:** Biodiversity is a valuable resource for scientific research, helping us understand ecological processes, genetics, and evolution.
v. **Aesthetic and Cultural Value:** Biodiversity enhances the beauty of the natural world and has cultural significance for many societies.
vi. **Ethical and Moral Reasons:** Many people believe that we have a moral and ethical responsibility to protect other species and maintain the diversity of life on Earth.

viii) Differentiate between renewable and non-renewable resources.

ANS: DIFFERENTIATE BETWEEN RENEWABLE AND NON-RENEWABLE RESOURCES:

ASPECT	RENEWABLE RESOURCES	NON-RENEWABLE RESOURCES
DEFINITION	Resources that can be naturally replenished or regenerated within a relatively short time frame (e.g., solar energy, wind energy, biomass).	Resources that are finite and cannot be readily replaced on a human timescale (e.g., fossil fuels like coal, oil, and natural gas).
AVAILABILITY	Can be sustained with responsible management and use.	Depletes over time and is exhaustible.
ENVIRONMENTAL IMPACT	Generally have a lower environmental impact and reduced greenhouse gas emissions.	Often associated with environmental degradation, pollution, and greenhouse gas emissions.
EXAMPLE	Solar energy, wind energy, hydropower, sustainable forestry, and agricultural crops.	Fossil fuels (coal, oil, natural gas), uranium (for nuclear energy), minerals (e.g., metals), and non-renewable groundwater.

CHAPTER: 26 **BIOTECHNOLOGY**

ENCIRCLE THE CORRECT CHOICE.

i) Cloning of genes creates an identical copy of a DNA sequence. The process of cloning is significant for what reason?

(a) To create a transgenic organism (b) To diagnose genetic disorders
(c) To study gene function (d) All the above

ii) DNA sequencing is a technique used to determine the sequence of nucleotides in a DNA molecule. What is the significance of DNA sequencing?

(a) To identify genetic variations (b) To create a transgenic organism
(c) To study gene expression (d) All the above

iii) DNA analysis is used to identify genetic disorders. What is the purpose of DNA analysis?

(a) Sequence of nucleotides in a DNA (b) To study gene expression
(c) To identify genetic variations (d) All the above

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

iv) Biotechnology has many applications in healthcare, including gene therapy, drug production, and diagnosis of diseases. What is the significance of biotechnology in healthcare?

(a) To study gene function (b) To improve animal welfare
(c) To develop new materials (d) All the above

v) Polymerase chain reaction (PCR) is a technique used to amplify DNA. What is the significance of PCR?

(a) To sequence DNA (b) To identify genetic disorders
(c) To study gene expression (d) All the above

vi) Gel electrophoresis is a technique used to separate DNA fragments based on size. What is the significance of gel electrophoresis?

(a) To study gene expression (b) To sequence DNA
(c) To identify genetic variations (d) All the above

vii) Sanger sequencing is a technique used to read the sequence of DNA. What is the significance of Sanger sequencing?

(a) To identify genetic disorders (b) To sequence DNA (c) To study gene expression (d) All the above

viii) Southern blotting is a technique used to transfer DNA fragments from a gel to a membrane. What is the significance of Southern blotting?

(a) To identify genetic disorders (b) To sequence DNA(c) To study gene expression (d) All the above

ix) Genomic maps are useful for:

(a) Identifying genes associated with specific diseases (b) Creating genetically modified organisms
(c) Analyzing DNA methylation patterns (d) Detecting protein-protein interactions

x) What is the significance of monoclonal antibodies in biotechnology?

(a) To study gene expression (b) Used to diagnose diseases
(c) Genetically modified organisms (d) They can be used to create new drugs

Q. no:	Answer	Q. no:	Answer
i)	(d) All the above	vi)	(d) All the above
ii)	(d) All the above	vii)	(b) To sequence DNA
iii)	(d) All the above	viii)	(a) To identify genetic disorders
iv)	(a) To study gene function	ix)	(a) Identifying genes associated with specific diseases
v)	(d) All the above	x)	(d) They can be used to create new drugs

WRITE SHORT ANSWER OF THE FOLLOWING:

i) Why Amp and Lac² are used in construction of rDNA?

ANS: Amp (ampicillin) and Lac² (lactose operon) are used in the construction of recombinant DNA (rDNA) for several reasons:

- Ampicillin is an antibiotic that can be used as a selectable marker in plasmids. Plasmids are small, circular pieces of DNA that can be used to carry foreign DNA (such as a gene of interest) into a host cell. By including an ampicillin resistance gene in the plasmid, only those host cells that have successfully taken up the plasmid (including the foreign DNA) will be able to grow in the presence of ampicillin. This allows for the selection of cells that have successfully undergone genetic transformation.
- Lac², or the lactose operon, is often used as a regulatory element in recombinant DNA constructs. The lac operon includes a promoter region that can be used to control the expression of genes of interest. By placing the gene of interest under the control of the lac promoter, researchers can regulate when and how the gene is expressed in the host cell.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

ii) What do you mean by RFLPS?

ANS: RFLPs, or Restriction Fragment Length Polymorphisms, are variations in the DNA sequences of individuals that result in differences in the lengths of DNA fragments produced by restriction enzyme digestion. RFLPs are used as genetic markers in various applications, such as DNA fingerprinting and genetic mapping. They are particularly useful in DNA profiling and forensic science to identify individuals or establish familial relationships.

iii) How is biotechnology used in healthcare, and what are some of the potential benefits and challenges associated with its use?

ANS: Biotechnology is used in healthcare in various ways, including:

i. **Pharmaceuticals:** Biotechnology is used to develop and produce biopharmaceuticals like insulin, vaccines, and monoclonal antibodies for treating diseases.
ii. **Genetic Testing:** Biotechnology enables the identification of genetic disorders, predicting disease risk, and providing personalized medicine options.
iii. **Gene Therapy:** Biotechnology allows the modification of genes to treat or prevent genetic diseases.
iv. **Diagnostics:** Biotech tools like PCR (Polymerase Chain Reaction) and DNA sequencing are vital for diagnosing diseases, including infectious diseases.
v. **Stem Cell Therapy:** Biotechnology is involved in developing stem cell-based therapies for various conditions.

iv) Why restriction enzymes are called molecular scissors?

ANS: Restriction enzymes are called "molecular scissors" because they cleave DNA at specific recognition sequences, creating double-stranded breaks in the DNA molecule. These enzymes recognize and cut at specific nucleotide sequences, often creating sticky ends or blunt ends. This ability to precisely cut DNA at specific locations is similar to how scissors cut paper, making them essential tools in genetic engineering and recombinant DNA technology.

v) What do you mean by palindrome?

ANS: In molecular biology, a palindrome refers to a sequence of DNA, RNA, or protein that reads the same forwards and backwards. In the context of DNA, palindromic sequences are essential for the recognition and cleavage of DNA by restriction enzymes. These sequences have two-fold rotational symmetry, making them suitable recognition sites for enzymes that cut DNA.

vi) Enlist enzymes use in rDNA technology?

ANS: Enzymes used in rDNA technology include:

i. **Restriction Enzymes:** These enzymes cleave DNA at specific recognition sequences, facilitating the insertion of foreign DNA into a vector.
ii. **DNA Ligase:** DNA ligase joins DNA fragments by catalyzing the formation of phosphodiester bonds between their ends.
iii. **Polymerases:** DNA polymerases like Taq polymerase are used for DNA replication and PCR.
iv. **Reverse Transcriptase:** This enzyme synthesizes DNA from RNA templates and is used in creating complementary DNA (cDNA) from mRNA.
v. **Endonucleases:** Endonucleases are used to cut DNA internally at specific sites, often for targeted gene disruption or modification.

CHAPTER: 27 — **BIOLOGY AND HUMAN WALFARE**

ENCIRCLE THE CORRECT ANSWER:

i) DT vaccine is used for
(a) Hepatitis (b) Measles (c) Common cold (d) Diphtheria toxoid

ii) Clostridium tetani which caused tetanus is a
(a)Virus (b) Bacteria (c) Parasite (d) Fungi

iii) MMR vaccine is effective for
(a) Measles (b) Polio (c) Measles, Mumps & rubella (d) Hepatitis

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

iv) The top milk producing animal of Pakistan is
(a) Bhadawari (b) Jaffrabadi (c) Nili- Ravi (d) Godavari

v) The yeast that gives puff appearance to dough is called:
(a) Saccharomyces (b) Lactobacillus (c) Penicillium (d) Escherichia coli

vi) The removal of floating solids and organic materials from the sewage is
(a) Primary treatment (b) Secondary treatment (c) Tertiary treatment (d) none of these

vii) The process of hybrid crossing with its parents is called:
(a) Selection (b) Hybridization (c) Back cross (d) Acclimatization

viii) French beans and Cauliflowers are vegetables of:
(a) Winter (b) Summer seasons (c) Spring (d) all

ix) The Science that deals with the modification of DNA and Enhances the characteristic of an organism is called
(a) Genetic Engineering (b) Biotechnology (c) Biology (d) None of these

x) Sludge digestion is a biological process in which organic solids are decomposed into stable substances
(a) Stable substances (b) non stable substances (c) Effluent (d) None of these

Q. no:	Answer	Q. no:	Answer
i)	(d) Diphtheria toxoid	vi)	(a) Primary treatment
ii)	(b) Bacteria	vii)	(b) Hybridization
iii)	(c) Measles, Mumps & rubella	viii)	(c) Spring
iv)	(c) Nili-Ravi	ix)	(a) Genetic Engineering
v)	(a) Saccharomyces	x)	(a) Stable substances

WRITE SHORT ANSWER OF THE FOLLOWING:

i) Why vaccination is important for infectious disease?

ANS: Vaccination is important for infectious diseases for several reasons:

i. **Disease Prevention:** Vaccines stimulate the immune system to produce antibodies and memory cells against specific pathogens. This prepares the body to recognize and fight the pathogen if encountered in the future, preventing infection or reducing the severity of the disease.
ii. **Herd Immunity:** Widespread vaccination in a population creates herd immunity, protecting even those who cannot be vaccinated, such as individuals with certain medical conditions or weakened immune systems.
iii. **Disease Eradication:** Vaccination has led to the eradication of some deadly diseases, like smallpox, and has brought others, like polio, close to eradication.
iv. **Cost-Effective:** Vaccination is often more cost-effective than treating diseases, as it reduces healthcare costs and lost productivity.
v. **Public Health:** Vaccination is a crucial public health measure, helping to control the spread of infectious diseases and protect communities.

ii) List the outstanding milk producing breeds of cow.

ANS: Some outstanding milk-producing breeds of cows include:

i. **Holstein-Friesian:** Known for high milk production, especially in temperate climates.
ii. **Jersey:** Known for high butterfat content in milk.
iii. **Guernsey:** Produces rich and creamy milk with moderate milk production.
iv. **Ayrshire:** Known for adaptability to various climates and good milk production.
v. **Brown Swiss:** Produces high-quality milk and is known for longevity.
vi. **Milking Shorthorn:** Combines good milk production with versatility.

SOLVE BOOK EXERCISE MCQs AND SHORT QUESTION ANSWERS

iii) What do you mean by integrated diseases and its management?

ANS: Integrated disease management refers to a holistic approach that combines various strategies to control and prevent diseases in agriculture, particularly in crops. It involves the integration of cultural, biological, chemical, and other control methods to minimize disease impact. The goal is to reduce the reliance on a single control measure and promote sustainability. Integrated disease management strategies may include crop rotation, use of disease-resistant varieties, biological control agents, fungicides, and cultural practices like sanitation.

iv) What is the role of livestock?

ANS: The role of livestock includes:

i. **Food Production:** Livestock provide meat, milk, eggs, and other animal-based products for human consumption.
ii. **Economic Livelihood:** Livestock farming is a source of income and livelihood for many people, especially in rural areas.
iii. **Agricultural Support:** Livestock can provide manure for fertilizing crops and can be used for plowing fields.
iv. **Diversity:** Livestock farming contributes to agricultural diversity, helping to reduce dependence on a single crop.
v. **Cultural and Traditional Value:** Livestock play a significant role in many cultures and traditions.
vi. **Biodiversity:** Certain livestock breeds are valuable for conserving genetic diversity within animal populations.

v) Differentiate between following:

(a) Selection and Backcross **(b) Vegetables of summer and vegetables of winter**

(c) Selection and Hybridization

ANS: DIFFERENTIATE BETWEEN:

(a) SELECTION AND BACKCROSS:

ASPECT	SELECTION	BACKCROSS
DEFINITION	Choosing individuals with desirable traits for breeding	Repeatedly crossing offspring with a parent with the desired trait
PURPOSE	Enhancing specific traits in a population	Introducing and amplifying a specific trait or gene
GENETIC DIVERSITY	Maintains genetic diversity	Reduces genetic diversity as it focuses on one trait
NUMBER OF GENERATIONS NEEDED	Typically requires multiple generations for improvement	Faster process, can achieve the desired trait in fewer generations

(b) VEGETABLES OF SUMMER AND VEGETABLES OF WINTER:

ASPECT	VEGETABLES OF SUMMER	VEGETABLES OF WINTER
PLANTING SEASON	Planted and harvested in warm months	Planted in cooler months, harvested in colder months
GROWTH CHARACTERISTICS	Rapid growth, warmth-loving plants	Slower growth, cold-hardy plants
EXAMPLES	Tomatoes, cucumbers, peppers	Carrots, broccoli, kale

(c) SELECTION AND HYBRIDIZATION:

ASPECT	SELECTION	HYBRIDIZATION
METHOD	Choosing individuals with desirable traits for breeding	Crossing two genetically different individuals
GENETIC DIVERSITY	Maintains genetic diversity	Combines genetic material from two different sources
INBREEDING	Helps reduce inbreeding by selecting within a population	May involve inbreeding to create pure lines before crossing
TIME AND COMPLEXITY	Generally simpler and quicker	May require more time and resources